ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN

NESOP (00115)

RQ-NESOP-S1

Intermediate Operator (Intrm Opr)

Developmental Period 2

Qualification Code: ALAZ

MITE ID: 120689

RECORD OF AMENDMENTS

The “Authority” reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with the Commander Naval Personnel and Training Group. The Curriculum Control Authority (CCA) for Chapters 3 and 4 is Canadian Forces Naval Operations and may be further delegated to the Standards Division Commander. The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J “Course History”.

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TABLE OF CONTENTS

CHAPTER 1: TRAINING MANAGEMENT DETAILS 1-4

AIM 1-1

COURSE DURATION CALCULATION 1-1

RESOURCES 1-1

COURSE CAPACITY 1-1

COURSE PREREQUISITES 1-1

RELATED DOCUMENTS 1-2

EVALUATION 1-2

VALIDATION 1-2

QUALIFICATION 1-2

PRIOR LEARNING ASSESSMENT 1-2

SPECIAL INSTRUCTIONS 1-2

CHAPTER 2: PERFORMANCE OBJECTIVES 2-1

PO 001 Generate Operational Messages 2-1
PO 002 Perform Electronic Warfare 2-2
PO 003 Operate EA Equipment 2-3

CHAPTER 3: ASSESSMENT OF TRAINEE 3-1

GENERAL 3-1

ENABLING CHECKS 3-1

PERFORMANCE OBJECTIVE WEIGHTING 3-1

PERFORMANCE CHECKS 3-1

PROGRESS MONITORING 3-2

UNSATISFACTORY TRAINING PROGRESS 3-2

UNSATISFACTORY CONDUCT 3-3

COURSE REPORTS 3-3

EVALUATION OF TRAINEE 3-3

RANKING 3-4

TOP STUDENT STANDING 3-4

TRAINING REVIEW BOARD 3-5

SPECIAL INSTRUCTIONS 3-5

CHAPTER 4: LESSON SPECIFICATIONS 4-1

PO 001 Generate Operational Messages 4-1
EO 001.01 Utilize Messages 4-1
EO 001.02 Utilize Message Drafting Software and Utilities 4-4
EO 001.03 Draft Operational Messages 4-6
Performance Check – PO 001 4-8

PO 002 Perform Electronic Warfare 4-9
EO 002.01 Intermediate Electromagnetic (EM) Theory 4-9
EO 002.02 Record Electronic Emissions 4-12
Performance Check – PO 002 4-14
PO 003 Operate EA Equipment 4-15
EO 003.01 Electronic Deception 4-15
EO 003.02 Direct EA Decoy Loading Procedures 4-18
Performance Check – PO 003 4-20

ANNEX A – MAIN REFERENCES A-1

ANNEX B – TASK LIST ..... B -1

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS C-1

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR D-1

ANNEX E – ACRONYMS / ABBREVIATIONS E-1

ANNEX F – SUGGESTED SEQUENCING AND TIMINGS F-1

ANNEX G – TRAINING RESOURCES G-1

ANNEX H – COSTING INFORMATION H-1

ANNEX I – WEEKLY STUDENT WRITE UP H-1

ANNEX J – COURSE HISTORY J-1

ANNEX K – EVALUATTION MATRIX K-1

FOREWORD

1. This training document is developed under the authority of the C Naval and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education; and
   b. A-PD-055/PP-002 Job Based Specification (NCM) Naval Electronic Sensor Operator (Draft) dated Jun 08.

2. This publication is effective on receipt and supersedes all previous versions. Performance Objectives (POs) are explained fully in Chapter 2. The Lesson Specifications (LSs)/Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). QSP revision procedures are outlined in the NTI. Amendments to this QSP may only be approved by the Commandant of CFNOS. Authority to approve amendments that affect chapters 3 and 4 may be delegated to the Standards Division Commander.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP

AIM

1. The aim of the NESOP RQ-S1 course is to train the NES Intermediate Operator in the operation of the Close In Weapon System (CIWS) and Electronic Warfare (EW) systems.

COURSE DURATION CALCULATION

2.

Actual Training Days (ATT) (Periods)	163
Training Administrative Time (TAT) / PT (IAW CDS Direction 2203.4) (Periods)	24
Base Administration Time (BAT) (Days)	2
Total Training Days	26

RESOURCES

3. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H. (add extra lines to table as necessary)

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Snr Instructor	PO1	AEEK, AHCJ	1	AIMU, AIUQ
OPI	PO2/MS	AAXO/ALBB, AIMU, AIUQ	1	AHCJ
Instructor	PO2/MS	AIXT/ALBA, AHCH, AIUQ	2	

CRITICAL TRAINING RESOURCES	**NO. OF DAYS**
Classroom	17.25
Naval Operational Trainers	2.75

COURSE CAPACITY

4. Planning for the conduct of training is based on the following guidelines:
   a. Maximum – 12;
   b. Minimum – 6; and
   c. Optimum – 10.

COURSE PREREQUISITES

5. In order to be selected for this training member shall:
   a. Be AIXA/ALAY qualified;
   b. Have completed one year post QL 4 employed; and

RELATED DOCUMENTS

6. NIL

EVALUATION

7. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques.

VALIDATION

8. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

9. Successful completion of this course constitutes the primary means of attaining the NESOPIntrm Opr NESOP RQ-S1 (Competency Code ALAZ/MITE).

PRIOR LEARNING ASSESSMENT

10. Prior Learning Assessment shall be conducted in accordance with A-P9-050-000/PT-012 Canadian Forces Military Equivalencies Program and NTI 005.

SPECIAL INSTRUCTIONS

11. NIL

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP

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PO 001 Generate Operational Messages

1. Performance: Generate Operational Messages

2. Conditions:

   a. Given:

      1) Message Drafting Software
      2) Work Station/PC
      3) References
      4) Supervision

   b. Denied: Nil

   c. Environmental:

      1) At sea / Alongside Day or Night
      2) Trainer/ Simulator Day or Night

3. Standard. IAW references and due regard for procedures, the NES Intermediate Operator shall generate operational messages under constant supervision to include:

   a. Utilizing message drafting software and utilities;
   b. Drafting operational messages by:

      1) Preparing Electronic Warfare OCS Intentions Messages
      2) Extract information from operations orders
      3) Provide input to various reports
      4) Extracting information from frequency plans

4. References.

A3, A9, A15, A22, A23, A25, A58, B8, B9, B12

5. Specification Task Numbers.

AT0016, BT0001, BT0002, BT0041, BT0042, BT0045, ET0002

6. Supporting Skills.

N/A

7. Supporting Knowledge.

BK0002, BK0004, BK0007, BK0008, BK0017, BK0059, BK0060, EK0002, EK0003, EK0004, EK0007

8. Training Limitations.

Nil

PO 002 Perform Electronic Warfare

1. Performance: Perform Electronic Warfare

2. Conditions:

   a. Given:

      1) Sea Search
      2) References
      3) Supervision

   b. Denied: Nil

   c. Environmental:

      1)At sea / Alongside Day or Night
      2)Trainer/ Simulator Day or Night

3. Standard: IAW references and due regard for safety procedures, the NES Intermediate Operator shall operate SSD for EW/AWW operations to include:

   a. Perform Intermediate Electromagnetic (EM) Theory
   b. Record Electronic Emissions

4. References.

   A30, A59, B2

5. Specification Task Numbers.

   BT0021, BT0022, BT0023

6. Supporting Skills.

   N/A

7. Supporting Knowledge.

BK0016, BK0017, BK0019, BK0022 BK0023, BK0024, BK0025, BK0027, BK0029, BK0030

8. Training Limitations.

   Nil

PO 003 Operate EA Equipment

1. Performance: Operate EA Equipment

2. Conditions:

   a. Given:

      1) References
      2) Supervision
      3) EW Equipment
      4) Calculator
      5) EW Software and applications
      6) Work Station/PC
      7) Munitions and Training Decoys

   b. Denied:

      Nil

   c. Environmental:

      1) At sea / Alongside Day or Night
      2) Trainer/ Simulator Day or Night

3. Standard: IAW references and due regard for safety procedures, the NES Intermediate Operator shall operate EA equipment to include:

   a. Performing Electronic Deception to include:

      1) Operating EA systems under constant supervision;
      2) Flashing up/shutting down EA systems under constant supervision;
      3) Conducting EA systems operational functional checks under constant supervision.

   b. Directing EA decoy loading procedures to include:

      1) Flashing up/shutting EA system under constant supervision; and
      2) Conducting EA systems operational functional checks under constant supervision.

4. References.

A2, A3, A4, A8, A14, A18, A19, A20, A21, A24, A32, A34, A39, , A45, A49, B9, B15, B16, B17, C5, C10, C11, C13, C15, C16, C17, C18, C20, C24

5. Specification Task Numbers.

CT0007, CT0012, CT0020, CT0030, CT0032, CT0033, CT0034, CT0039, CT0043, CT0045, CT0046, CT0048, CT0049, CT0050, CT0051, CT0053, CT0054, CT0055, CT0057, CT0058, CT0059, CT0060, CT0062, CT0061, CT0063, CT0064, CT0066, CT0067, CT0068, CT0069, CT0070, CT0071, CT0072, CT0074, CT0076, CT0077, CT0078, CT0079, CT0080, CT0081, CT0082, CT0083, ET0006, ET0008.

6. Supporting Skills.

Nil

7. Supporting Knowledge.

AK0006, CK0009, CK0011, AK0029, CK0034, CK0037, CK0044, CK0045, CK0047, CK0048, CK0049, CK0050, CK0051, CK0052, CK0053, CK0055, CK0056, CK0057, CK0058, CK0063, , CK0076, CK0077, CK0078, CK0079, CK0080, CK0081, CK0082, CK0083, CK0084, CK0085, CK0086, CK0087, CK0089, CK0090, CK0091, CK0092, CK0094, CK0095, CK0096, CK0097, CK0098, CK0099, CK0100, CK0101, CK0103, CK0104, CK0105, CK0106, CK0107, CK0108, CK0109, CK0110, CK0111, CK0112, CK0113, CK0115, CK0117, CK0118, CK0119, CK0120, CK0121, CK0122, CK0123, CK0124, CK0125, CK0126, CK0127, CK0128, CK0129, CK0131, CK0132, DK0030

8. Training Limitations.

Nil

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF CFNOS

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GENERAL

1. Trainees will be developed and assessed using a combination of formative and summative measurement instruments. At a minimum, these will involve Performance Checks (PCs), Critical-Enabling Checks, Enabling Checks (ECs), end of lesson confirmations, practice periods, homework, assignments and quizzes. See Annex K (Evaluation Matrix) for details. As well professionalism, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed as per paras 13 and 14. A trainee who displays inappropriate professionalism, ethics, demeanour and/or behaviour is considered to have demonstrated unsatisfactory conduct and will immediately result in the convening of a Training Review Board (TRB), for conduct, in order to assess the suitability of the trainee.

2. Students must achieve a pass in all PCs and Critical-ECs in order to achieve an overall passing mark in the NES Intermediate Operator NESOP RQ-LStraining and be granted competency and course ID associated with this training. This trainee assessment needs to be read in conjunction with EOs and the evaluation matrix in Annex K.

ENABLING CHECKS

3. ECs are designed to provide feedback to the instructor and trainee and shall be used to prepare the trainee for subsequent PCs and Critical-ECs.

4. The Pass/Fail criteria for all EC are 80%.

5. In the event of an EC failure, the trainee IAW TE Policy will be counselled and/or a Progress Review Board held, and given, as appropriate, additional:

a. Training
b. Assignments;
c. Practice periods; and/or
d. Studies.

PERFORMANCE OBJECTIVE WEIGHTING

To be used in final mark calculation based on Critical EC and PC's.

PO#	Name	Ranking	Weight (%)	PC Weight
001	Generate Operational Messages	3	30	30
002	Perform Electronic Warfare	1	40	40
003	Operate EA Equipment	2	30	30

PERFORMANCE CHECKS

6. PCs and Critical-ECs are the primary means of Summative assessment and usually occur after all Enabling Objectives (EOs) have been completed for a specific Performance Objective (PO). Summative tests determine whether learners have achieved POs, or critical EOs (those deemed prerequisite to further IT&E). Summative tests are used at the end of a phase of instruction.

7. PCs will normally be conducted after a series of formative testing has occurred and the student has had the opportunity to develop skills and apply knowledge related to the PO. For

NES Intermediate Operator NESOP RQ-LStraining, some PCs are divided in to both Practical PCs as well as Critical-ECs:

8. Critical-ECs identify key objectives or subject matter too small to count as a PC but are essential for further training and must be known at the NES Intermediate Operator NESOP RQ-S1 level. Critical-ECs are to be interpreted and treated the same as a PC regarding unsatisfactory trainee progress and convening of TRBs. The PC and Critical-ECs assess the knowledge or ability of the trainee to perform job related duties or assignments once onboard and are designed to reflect actual work situations as close as possible.

9. The Pass/Fail criteria for all PC and Critical-EC, and formative assessment tools such as EC, homework, quizzes, assignments, practical exercises is 80%.

10. All PC or Critical EC failures and all supplement PC or Critical EC failures are to be handled IAW NAVORD 4500-0, the direction provided in the QSP and TE policy

PROGRESS MONITORING

11. Trainee progress will be monitored using formative tests to help the learner and the instructor recognize progress or lapses in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are used during a phase of instruction. Formative testing (ECs, homework, assignments, quizzes, practice periods, etc.) and any method or tool used to develop the trainees shall not be used in the calculation of the students overall mark.

12. The formative type of development and testing shall be used to monitor the students and determine whether or not they are progressing. Formative testing identifies both strengths and weaknesses in the student's knowledge and performance of a given subject. When weaknesses are noted the instructors shall provide additional training prior to a PC. ECs shall not be formally retested but may be used to assist in developing weak areas.

13. Trainee progress shall be continually monitored and recorded to provide:

a. Feedback on effectiveness of training;
b. Early warning of trainee difficulties;
c. Information to the TRB; and
d. Information to assist in completion of the CF 377, Course Report.

14. The course OPI shall maintain, on file, the results of all:
a. PCs and Critical ECs;
b. all formative testing material, ECs, practice periods, homework, assignments and quizzes; and
c. Conduct (See Annex L Student Write Up)

UNSATISFACTORY TRAINING PROGRESS

15. Unsatisfactory training progress is identified when:

a. trainee fails to pass any Critical-EC or PC on the 1st attempt;
b. a trainee fails to achieve the standard during a supplemental PC or Critical-EC; and
c. a trainee fails a combination of any three (3) PCs or Critical-ECs.

16. IAW TE policy , trainees will be afforded applicable remedial training following a first PC or Critical-EC failure and will be permitted a supplemental PC and Critical-EC no sooner than 48 hours and no later than one (1) week from the first failure date. In optimum circumstances, supplemental PCs and Critical-EC should be administered 48 to 72 hours after the first failure. It the student fails the supplemental PC or Critical-EC, a TRB will be convened. The highest mark achievable in any Critical-EC or PC supplemental is 80%.

UNSATISFACTORY CONDUCT

17. Unsatisfactory conduct is identified by the trainee's attitude, display of inappropriate professionalism, ethics, demeanour and/or behaviour during the duration of the course and shall be recorded IAW paras 13 & 14.

COURSE REPORTS

9. A Form 377, Canadian Forces Course Report, shall be prepared and distributed IAW DAOD 5031-9, CMP Instructions 01/06, NTI 1/08 (Course Reporting and Certificates), NPTG 4500-51 two working days after course completion.

EVALUATION OF TRAINEE

10. Evaluation of trainee performance shall be based on the following criteria, IAW COURSE REPORTING AND CERTIFICATES NAVAL TRAINING INSTRUCTION (NTI) 1/08:

a. Pass – 80%, successfully completed all Performance Objectives;

b. Fail – unable to:

i. safely or successfully complete all POs; or
ii. maintain adequate progress within the time, resources and re-testing limits set forth in the QSP; or

c. Incomplete – unable to complete a course due to:

i. medical, personal or service reasons; or
ii. the inability of the TE to carry out all of the POs.

11. The narrative description section is for providing additional information on the trainee's performance during the course. Performance during formative assessments shall be captured in the narrative to reflect exceptional and/or unsatisfactory performance, strengths and weaknesses. It is not necessary to list the POs in the narrative description.

12. A narrative description is to be completed for trainees under the following circumstances:

a. trainee demonstrated outstanding performance on any PO;
b. trainee experienced difficulty achieving the standard in any PC or Critical EC;
c. exceptional or unsatisfactory formative evaluations;
d. any conduct, professionalism or ethical behaviour relevant to course performance; and/or
e. any conditions and situations that affected course performance (family death).

RANKING

13. With regards to completing CF 377, ranking shall not be used.(IAW NTI 1/08 Course Reporting and Certificates, NPTORD 4500-51 letter grading and/or ranking are only required in exceptional circumstances.)

TOP STUDENT STANDING

14. Annex L (Student Write up) shall be used to record student's non academic progress on a weekly basis and it will be completed by the current instructor and controlled by the course OPI. Weekly student write ups will be held on the student's file.

15. Overall standing for top student will be IAW the table below.

CRITERIA	VALUE	COMMENTS
Academics	95	Based on PCs & Critical ECs results
Duties of Class Senior	2.5	Based on checklist Note 1
Dress and Deportment	2.5	Based on checklist Note 2
Total	100	

Note 1:

Students can only lose a maximum of 2.5% of their total mark towards the Top Student award in this section or gain a maximum of 2.5%. All students are initially given 2.5%.

-.25%

a) more than 3 improper reporting procedures to instructors in 1 week;
b) failure to inform instructors of location of student/students;
c) failure to ensure classroom is secured at end of day; and
d) failure of following orders given by instructors.

-.50%

a) 2nd failure to inform instructors of location of student/students;
b) 2nd failure to ensure classroom is secured at end of day; and
c) subsequent failures of duties listed above.

+.50%

a) Week without incident as course senior

Note 2:

-.25% picked up during weekly morning inspection;
-.50% picked up for 2nd infraction of same item that was previously noted and not satisfactorily corrected;
-1.0% picked up on CFNOS Divisions;
+.50% students that for 3 consecutive weeks maintain the CFNOS requirement for standard of dress; and
+1.0% student recognized by inspecting staff for having outstanding uniform at CFNOS Divisions.

TRAINING REVIEW BOARD

16. TRB shall be convened IAW NAVORD 4500-0 and TE policy for unsatisfactory training progress when a trainee fails:

a. to achieve the standard (80%) during a supplemental PC or Critical-EC;
b. a combination of any three (3) separate PCs or Critical ECs; and
c. to display appropriate conduct. (IAW Annex L Student Write up)

17. After missing 2 consecutive training days, a TRB will be convened to assess the trainee's suitability to continue training.

18. When member's absence requires course removal for medical, compassionate or other non-academic reasons, the absence may be dealt with by an Administrative (paper) TRB.

SPECIAL INSTRUCTIONS

19. NIL

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF CFNOS

PO 001 Generate Operational Messages
EO 001.01 Utilize Messages

1. Performance: Utilize Messages

2. Conditions:

   a. Given:

      1) Message Drafting Software
      2) Work Station/PC
      3) References
      4) Supervision

   b. Denied: Nil

   c. Environmental:

      1) Classroom

3. Standard. IAW references and due regard for security procedures, the NES Intermediate Operator shall utilize Messages under normal supervision.

4. Time.

   a. Classroom 14 x 50 min periods
   b. Practical (Classroom) 1 x 50 min periods

5. Method / Media.

   a. Lecture
   b. Practical

6. References.

   a. A3, A9, A58, B8, B9, B12

7. Training Aids.

   a. PPT
   b. Message Drafting Software

8. Test Details. (See Annex K for details)

   a. Formative: Tested in end of lesson confirmation and quiz. No formal EC.
   b. Summative: Tested as part of practical PC 001. Pass = 80%

9. Remarks.

   a. During the lecture portion, the Instructor will employ guided discussion techniques to amplify reasons for employment of operational messages.

10. Teaching Points.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	Sub-Sub Teaching Point	C	D	P	Method / Media	Ref
Types of Messages	Operational Messages		3	1		Lecture / Practice	A4 B8 B9
	-Message(s) description		2				
	-Message Application		2				
Interpret Operational Orders	Op Order Formats	Coastal OP Orders (to include major warfare areas AA, SU, EW, Comms and OP GEN)	3			Lecture	
Message Format	Types of Exercise Messages (Relevant current operational messages)		1			Lecture	B12
	Message Distribution		1			Lecture	A3 B8
	Extract info from Frequency Plan		1			Lecture	A3 A58
	Message Filing procedures		1			Lecture	B8

EO 001.02 Utilize Message Drafting Software and Utilities

1. Performance: Utilize Message Drafting Software and Utilities to include TACELINT Generator Software.

2. Conditions:

   a. Given:

      1) Message Drafting Software
      2) Work Station/PC
      3) References
      4) Supervision

   b. Denied: Nil

   c. Environmental:
      1) Classroom

3. Standard. IAW references and due regard for security regulations, the NES Intermediate Operator Utilize Message Drafting Software and Utilities under constant supervision.

4. Time.

   a. Classroom: 1 x 50 min period
   b. Demo: 1 x 50 min period
   c. Practical: 2 x50 min period

5. Method / Media.

   a. Lecture
   b. Demo
   c. Practical

6. References.

   a. A3, A23

7. Training Aids.

   a. PPT
   b. Trainer

8. Test Details. (See Annex K for details)

   a. Formative: Tested in end of lesson confirmation, practice. No formal EC.
   b. Summative: Tested as part of practical PC 001. Pass = 80%

9. Remarks

   a. Nil

10. Teaching Points.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	C	D	P	Method / Media	Ref
Utilizing Message Drafting Software and Utilities to include TACELINT Generator Software	Message Drafting Software and Utilities	1	1	2	Lecture/ Demo/ Practical	A3 A23

EO 001.03 Draft Operational Messages

1. Performance: Draft operational Messages

2. Conditions:

   a. Given:

      1) Message Drafting Software including TACELINT Generator Software
      2) Work Station/PC
      3) References
      4) Supervision

   b. Denied: Nil

   c. Environmental:

      1) Classroom

3. Standard. IAW references and due regard for security regulations, the NES Intermediate Operator Draft Operational Messages under constant supervision.

4. Time.

   a. Classroom: 22 x 50 min periods
   b. PC 1x 50 min period

5. Method / Media

   a. Lecture
   b. Demonstration
   c. Practical

6. References.

   a. A15, A22, A25, B9, B12

7. Training Aids.

   a. PPT
   b. Classroom

8. Test Details.

   a. Formative: Tested in end of lesson confirmation and practice. No formal EC.
   b. Summative: Tested as part of practical PC 001. Pass = 80%

9. Remarks.

   a. PC 001 to include drafting the following messages: TACELINT and OCS Intentions messages. Member must utilize appropriate message drafting software Teaching Points.

b.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub teaching Point	Sub-Sub teaching Point	C	D	P	Method / Media	Ref
Prepare Messages			1			Lecture	A15A 22
	Formats for Exercise Intention Messages	OCS Int utilizing applicable coastal refs	1	1	10	Lecture	B12
	Utilize TACELINT Generator program		1	1	6	Lecture	A15 A22
Provide Input to various deployment reports			1			Lecture	A25A 30

Performance Check – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. AWW scenario.

2. PC / Laptop with message drafting software.

ARRANGEMENTS AND PROCEDURES

3. This PC an open book practical exam to test the trainee's ability to write an EWX message.

4. Time to complete one period.

5. The trainee shall complete this PC individually.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. This PC will test your knowledge of drafting EW correspondence and the ability to draft applicable messages.

2. This PC will shall be an open book practical exam to test your ability to write an EWX message.

4. You must achieve a mark of 80% to pass part one.

5. You must complete message format within the time allotted and be assessed as Pass / Fail.

6. You shall complete this PC individually

PO 002 Perform Electronic Warfare
EO 002.01 Intermediate Electromagnetic (EM) Theory

1. Performance: Utilize Electromagnetic (EM) Theory to perform Signal Analysis

2. Conditions:

   a. Given:
      1) Supervision;

      2) Classroom;

      3) Access to References; and

      4) PC / Laptop with appropriate software.

   b. Denied: Nil

   c. Environmental:
      1) Classroom

3. Standard. IAW applicable references, the trainee shall Utilize Electromagnetic (EM) Theory to perform Signal Analysis under normal supervision.

4. Time.

   a. Classroom 36 X 50 min periods
   b. EC 1 x 50 min period

5. Method / Media.

   a. Lecture

6. References.

   a. A30, A59,B2,

7. Training Aids.

   a. PPT
   b. Publications applicable to job
   c. Calculator

8. Test Details.

   a. Formative: Developed through end of lesson confirmation, and assessed in written theory based EC using practical questions. EC 002.01. Pass = 80%
   b. Summative: Assessed as part of practical PC 002. Pass = 80%

9. Remarks.

   N/A

10. Teaching Points.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	Sub-Sub Teaching Point	C	D	P	Method / Media	Ref
Electromagnetic (EM) Theory	EM Spectrum Terminology	Definitions to include but not limited to: Polarization; Optical Spectrum; Infra-Red; Electro-Optic; Radiant Energy; Radiant Power; Radiant Existence; Radiance; Short Wave; Long Wave; Atmospheric Transmission; Detectors(photon effects and thermal effects); lasers; fibre optics; Electro-optical systems; and harmonics	8			Lecture	C13
	Define terms associated with radar signals	Definitions to include but not limited to: Types of wave forms (sine wave, square wave, triangle wave, sawtooth and pulse wave)	5			Lecture	
	Describe concepts of radar modulation	To include Amplitude, Frequency and Phase Modulation	4			Lecture	
	Continuous wave	Advantages, limitations and	3		magnetron;		
Radar Components	Define radar components functions and limitations	Doppler effect Timer; Transmitter; Antenna; wave guide; duplexer; amplifier; receiver; detector; power supply; wave guide dryer; feed horn; modulator;	8				

Lecture

Lecture

		klystron; travelling wave tube;					
Types of radars	To include but not limited to: Synthetic aperture Radar; three Dimensional; phases array, MTI radar and pulse doppler	MTI radars to include coherent, non-coherent and blind speed	4				
Describe Parametric Values under normal supervision	Terms and definitions associated with radar range equation	Range resolution; Radar range; minimum range; maximum unambiguous range; plumbing loss; beam shaping loss; collapsing loss; tube life loss; operator loss and field degradation loss	4				

EO 002.02 Record Electronic Emissions

1. Performance: Perform ELINT Collection Supervision

2. Conditions:

   a. Given:
      1) References
      2) Supervision
      3) EW Equipment
      4) EW Software and Applications
      5) Work Station/PC

   b. Denied: Nil

   c. Environmental:

      1) Classroom

3. Standard. IAW references and due for security procedures, the NES Intermediate Operator shall supervise electronic recording independently.

4. Time.

   a. Lecture 3 x 50 min periods
   b. Practical 13 x 50 min periods
   c. EC 8 x 50 min periods

5. Method / Media.

   a. Demo
   b. Practical

6. References.

   a. A30

7. Training Aids.

   a. Trainer

8. Test Details.

   a. Formative: Tested through end of lesson confirmation, homework and covered as part of Practical EC 002.03. Pass = 80%
   b. Summative: Assessed as part of practical PC 002. Pass = 80%

9. Remarks.

   a. N/A
      Teaching Points.

b.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	Sub-Sub Teaching Point	C	D	P	Method / Media	Ref
Priority of Collects	NEWC priorities of SEI and TACELINTS		1				A30
	SEI collect procedures		2		13		
	Quality assessment of collects	Rejections					
		Collect considerations					
	Understanding USSID 351	PRI classification of pulse type radar					

Performance Check – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom

2. Stationary

ARRANGEMENTS AND PROCEDURES

3. PC002 will be a Written test based on content from EO002.01 and EO002.02.

4. Pass mark will be 80%

5. Time Required 1 period.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC002 will be a Written test based on content from EO002.01 and EO002.02.

2. You must complete this PC individually and without assistance. References are not permitted.

3. Pass mark is 80%

4. You will have 1 x 50 min period to complete this exam.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. NIL

PO 003 Operate EA Equipment
EO 003.01 Electronic Deception

1. Performance: Utilize Electronic Deception to perform EA Attack

2. Conditions:
   a. Given:
   b. Given:
      1) References
      2) Supervision
      3) Classroom
      4) PC/ Laptop with appropriate software

   c. Denied: Nil

   d. Environmental:

      1) Classroom

3. Standard. IAW references and due regard for safety procedures, the NES Intermediate Operator shall utilize electronic deception theory.

4. Time.

   a. Classroom: 14 x 50 min periods
   b. EC: 1 x 50 min period

5. Method / Media.

   a. Interactive Lecture
   b. Lecture

6. References.

   a. A2, A3, A4, A18, A38, B9, C15, C16, C17

7. Training Aids.

   a. PPT

8. Test Details.

   a. Formative: Tested through end of lesson confirmation and in practical EC 003.01. Pass = 80%
   b. Summative: Assessed as part of practical PC 003. Pass = 80%

9. Remarks.

a. N/A
b.

10. Teaching Points.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	Sub-Sub Teaching Point	C	D	P	Method / Media	Ref
Jamming Theory	Jamming Definitions	To include but not limited to: Active jamming; self screening; escort screening; stand off; and forward deployed	2			Interactive Lecture	A2
	-Jamming Techniques	Common Techniques to include but not limited to: Frequency and polarity match; continuous interference; signal to noise ratio; and enemy EPM				Lecture	
	-Burn through concept					Lecture	
	Radar Cross Section	RCS reduction	3			Lecture	A4
	-Types of Noise Jamming	To include: Spot; Swept Spot; Barrage noise; and Sidelobe Jamming					
	-Deception Jammers						
	-Range Gate Pull-off						
	-Protective Measures	4 Basic types: Frequency Control; Antenna Design; Receiver Modification; and Processor Logic Circuits	2			Lecture	C16
	Disadvantages of Jamming	;To include but not limited to: Give Away Presence; HOJ; DF; and Interference	3			Lecture	B9
	IR Jamming		1			Lecture	A18
	Communications		1			Lecture	A18

	Jamming						
Electronic Attack	Role of Electronic Attack		1			Lecture	A3A 18A 38
	Types of EA	To include but not limited to: Radar Noise Jamming; Infra-Red Jamming; Communications Jamming; Electronic False Targets; and Break-Lock Devices	1			Lecture	A3A 18A 38

EO 003.02 Direct EA Decoy Loading Procedures

1. Performance: Direct EA Decoy Loading Procedures

2. Conditions:

   a. Given:

      1) References
      2) Supervision
      3) EA Equipment
      4) Dummy Rocket and Test Probe

   b. Denied: Nil

   c. Environmental:

      1) Classroom
      2) Trainer

3. Standard. IAW references and due regard for safety procedures, the NES Intermediate Operator shall Direct EA Decoy Loading Procedures.

4. Time.

   a. Classroom 13 x 50 min periods
   b. Demo 2 x 50 min periods
   c. Trainer 8 x 50 min periods
   d. EC: 8 x 50 min periods
   e. PC 1 x 50 min period per student

5. Method / Media.

   a. Lecture
   b. Practical

6. References.

   a. A2, A4, A8, A14, A24, A45

7. Training Aids.

   a. PPT
   b. Trainer

8. Test Details.

   a. Formative: Tested through end of lesson confirmation.
   b. Summative: Assessed as part of practical PC 003. Pass = 80%

9. Remarks.

   N/A

10. Teaching Points.

C=Classroom, D = Instructor Demo, P = Student Practical time (Trainer or Class)

Main Teaching Point	Sub Teaching Point	C	D	P	Method / Media	Ref
EA Decoy systems CAPS and LIMS under normal supervision	EA Decoy system Modes of Operations	1			Lecture	A2 A4 A24
	EA Decoy system operating Procedures	1			Lecture	A24
Flash-up/ Shut Down EA System under constant supervision	EA Decoy Systems Flash-up/Shut Down Procedures	1	.5	4	Lecture/ Demo/ Practical	A24
Conduct EA Operational Functional Checks under constant supervision	EA Decoy Systems functional Checks	1	.5	2	Lecture/Demo / Practical	A24
EA Decoy systems Drill and Procedures independently	EA Decoy Loading Procedures	1	1	2	Lecture/ Demo/ Practical	A24
	EA Decoy safety Regulations	1			Lecture	A8 A14A24
HCM OverView to include Major Components, Capabilities and Limitations.	57mm MKIII CEROS 200 MASS Loader Elisra Smart 3D Radar CMS	7			Lecture	XX

Performance Check – PO 003

PART 1 – ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Operational Shield II System.

ARRANGEMENTS AND PROCEDURES

2. All materials to be in place prior to commencement of PC, also chaff magazine is to be open.

3. Steps 2, 3, 4, 5, 11 and 16 on the checklist are critical, missing anyone of these steps will constitute a **UNSAT** on this PC.

4. The Student will complete this PC individually.

5. The Student will be permitted references except drill and procedures manual during live firing.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. N/A

PART III – PERFORMANCE CHECK NOTES

1. Test Specifications: IAW Part IV Performance Checklist.

PART IV – PERFORMANCE CHECK LIST

Name:_____ Course: _____
Date:_____ Administrator: _____

ACTION	REMARKS	SAT	UNSAT
1. Closing – up and clearing away.	- Ensure Chaff Crew Mustered. - Checks Action dress - Draw Keys to appropriate bridge module and loader units IAW CPF weapons key organization. - Establishes and checks comms with EWS and magazine. - Switches loader units to live. - Ensures SHIELD Bridge Module is set to EWD. -Ensures Signs posted. - Ensures proper closing up and clearing away of SHIELD launchers/magazine is carried out by the loaders. - Ensures the launcher deck areas are clear of crew and unnecessary personnel. - Receives report from magazine. - Reports "Port/Stbd SHIELD closed up and cleared away."		
Question #1:			
2. Receives Policy and loading instructions (audible alarm or verbally) switches to **local** on the bridge module and **safe** on the loader units.			
3. Requests permission from OOW for loaders to proceed to the launchers (ensuring **RADHAZ safety** conditions have been met)			
4. Ensures Team has been given the **Safety Brief**. - Ammunition markings and colour coding; - Ammunition handling; - Personnel safety; - Efflux zones;			
- Load out IAW policy order; 5. Ensures loaders load SHIELD and correct munitions **safely.**			
6. Ensures loaders proceed clear of launcher area (inside bridge, bridge lobby, or magazine as briefed).			
7. Switches loader units to live and bridge module to EWD and reports " SHIELD standing to".			
Question #2:			
LOCAL CHAFF/IR AND MISFIRE FIRING		**SAT**	**UNSAT**

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Version No:2.7
MITE Crse ID:120689

8. Switched Bridge Module EWD / LOCAL toggle switch to LOCAL. a. Observe EWD / LOCAL indicator. b. Observe Load Status.			
9. Press IR / CHAFF button to select CHAFF.			
10. Presses DIS or SED.			
11. Pressed Local ALARM switch. (3 Times)			
12. Pressed FIRE FORWARD or AFT to select firing direction. c. Observe Barrel Status for round fired. d. Makes proper voice report.			
13. Switches Bridge Module EWD / LOCAL toggle switch to LOCAL. e. Observe EWD / LOCAL indictor. f. Observe Load Status.			
14. Pressed IR or CHAFF switch.			
15. Pressed DIS or SED.			
16. Pressed Local ALARM switch. (3 Times)			
17. Pressed FIRE FORWARD or AFT to select firing direction. g. Observe bridge module flashing light indicator. h. Make proper Misfire report. i. Wait for orders from SWC. j. Select SAFE. k. Brief loaders and send them out. l. Loaders back, select LIVE + EWD. m. Make proper report.			
Question #3:			
~~18. Completed Task within 30 Minutes.~~			
~~19. Observed all Safety Precautions.~~			

OVERALL ASSESSMENT			
Administrator Comments			
ACTION	**REMARKS**	SAT	UNSAT
Administrator Signature:	**Trainee Signature:**		

ANNEX A – MAIN REFERENCES

A = CF or official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other publication

A1	CFCD 102 Maritime Command Combat Readiness Requirements
A2	CFCD 106 Maritime Tactical Orders
A3	CFCD 107 Maritime Command Communications Orders
A4	CFCD 114 Canadian Maritime EW Orders
A5	MARCOD 43-02 Radio Frequency (RF) Hazards (RADHAZ) Safety Policy and Procedures
A6	MARCORD 46-01 Missile Firing Policy and Process
A7	MARCORD 46-02 Weapons Systems Certification Policy
A8	MARCORD 46-03 Safety Firing Orders for Ships and Submarines
A9	MARCORD 46-04 Weapons Keys Policy for Ships
A10	NCPM 701 Halifax Class AAW Drills and Procedures Manual
A11	C-09-153/TS-000 Explosives Safety Manual
A12	C-74-300-A01/Nj-000 Identification of Ammunition
A13	CFCD 111 Canadian Standards Operating Procedures & Guidance Manual for Maritime Tactical Data Link
A14	C-74-315-NAD/TA-000 Ammunition Shield II Decoy 100mm
A15	MARLANTORD 44-4 Requests for Service Support
A16	C-17-912-000/Mm-001 Technical Maintenance
A17	Sea Training Guide
A18	National Defence Security Instructions (NDSI)
A19	NEWC Secure Website
A20	C-58-251-000/MB-001 ESM Set AN/SLQ-501
A21	C-58-251-000/MB-201 System Operation (CANEWS)
A22	MARPACORD 45-8 Request for Service Support
A23	DND Message Drafting Software
A24	C-58-258-000/MS-003 Shield II Type 200 Chaff Decoy
A25	MARCORD 71-11 Post Deployment Reporting (PDR)
A26	B-GT-D35-001/AG-000 Management of the Radio Frequency Spectrum
A27	
A28	C-06-050-029/AD-000 Antenna and Radio Propagation
A29	CF1497 Maritime Intercept Report
A30	MARLANTORD 71011 Post Deployment Reports
A31	Halifax Class Combat System Software (CCS) Document
A32	MARCORD 59-1 Guidance for the Deployment of Portable Intelligence Systems
A33	NOSTP 200
A34	Trinity Website
A35	Secure MIPS Website
A36	Ship's Standing Orders
A37	Direction Finder Set AN/SRD-504
A38	MCION Secure Website
A39	C-70-374-000/MS-002 Mk 15 Close Weapon System
A40	Electronic Warfare Support Software (EWSS) Removable Hard-Drive
A41	Canadian Forces Electronic Warfare Data Base (CFEWDB)
A42	Naval Intercept Data Base (NEDB)
A43	NEWC USSID 351 Handbook
A44	Naval Electronic Warfare Data Base (NEWDB)

A45	RAMSES Theory of Operations User Manual C-58-259-000/MD-2001 Books 1-5
A46	RAMSES Jam Program Listing
A47	Sea Search Hard Drives Bay 1/2
A48	NULKA Active Decoy System C-58-371-000?MB-001
A49	Sea Search SOP and Secret Supp
A50	B-G G-005-004/DF-005 Use of Force
A51	MAOP 331 Handbook of the Law of Naval Operations
A56	Communications ESM Handbook
A57	Pro User Guide
A58	MARCORD 48-05 Frequency Management Policy
A59	MARCORD 48-03 Naval Tactical data Link Policy
A60	
B1	ACP 125
B2	ADATP 1.1 (Link)
B3	AKAC 1553
B4	AMSC TC 609
B6	AMSC TC 1800
B7	AMSH 1707
B8	APP 11 NATO Message Catalogue
B9	ATP 1 (Vol 1) Allied Maritime tactical Instructions & Procedures
B10	ATP-29 NATO Maritime Unit data & Equipment List
B11	ATP-31 NATO Above Water Warfare Manual
B12	AXP-2 Allied Exercise Publication
B13	AXP 5 NATO Experimental Tactics & Amplifying Tactical Instructions
B14	AXP 5(SE Supp) Secret Extrac Supp To NATO Experimental Tactics & Amplifying Instructions
B15	EA 2810 Fundamentals of Tech ELINT
B16	Math for SIGINT (1030 Math Package)
B17	NWP 10-1-40 Electronic Warfare Coordination
B18	NWP 10-1-12 Maritime Reporting System
B19	Anti-Air Missile Support
B20	Anti-Air Cruise Missile
B21	
B22	ATP-29 Supp 1 NATO Maritime Equipment Capabilities and Data
B23	ATP-1 (Vol 2) Allied Maritime Tactical Signal & Manoeuvring Book
C1	Jane's Publication
C2	Applied ECM Volume 1 & 2
C3	Electronic Countermeasures
C4	Principles of Naval Weapon system
C5	AN/ULR 501 Acquisition Detection System Operators Manual
C6	Block 1B CIWS Handbook
C7	
C8	NAVSEA ULSS-15/1B Operational Logistic Support
C9	
C10	Applied ECM Vol 1
C11	Applied ECM Vol 2
C12	Combat Fleets of the World
C13	Introduction to Radar Systems
C14	Lloyd's Registry
C15	Principles of Radar
C16	Radar Fundamentals
C17	Radar Handbook
C18	Radar System Analyses

C19	AN/SRd-504 Factory Training Guide

C20	Complete Idiot's Guide to Unix
C21	SIRIUS Technical manual
C22	Phalanx instruction Manual (PIMS)
C23	CPF Documentation
C24	Unix Guide for Dummies
C25	SG-150 Operator Handbook
C26	AN/SPS-49 Operating manual

ANNEX B – TASK LIST

		PO	QL5 Substantiation
AT0001	Conduct communication terminal functional checks	N/A	N/T NCH
AT0002	Configure communication terminal for operation	N/A	N/T NCH
AT0003	Conduct communication checks	N/A	N/T NCH
AT0004	Maintain call sign lists	N/A	N/T QL4 SDP
AT0005	Extract information from COMPLANS,	N/A	N/T
AT0006	Communicate internally on communication equipment	N/A	N/T QL4 SDP
AT0007	Communicate externally on communication equipment	N/A	N/T QL4 SDP
AT0008	Maintain communication security	N/A	N/T QL4 SDP
AT0009	Utilize tactical voice procedures	N/A	N/T QL4 SDP
AT0010	Monitor tactical voice procedures	N/A	N/T QL4 SDP
AT0011	Utilize tactical codewords	N/A	N/T QL4 SDP
AT0012	Conduct Beadwindow procedures	N/A	N/T QL4 SDP
AT0013	Authenticate communication transmissions	N/A	N/T QL4 SDP
AT0014	Conduct anti-jam voice procedures	N/A	N/T QL4 SDP
AT0015	Extract information from EMCON Plans	N/A	N/T
AT0016	Extract information from Frequency Plans	001	Train
BT0001	Provide input to various Deployment reports	001	Train
BT0002	Provide input to Operation Orders (OPORD)	001	Train
BT0005	Complete operational formex's	N/A	N/T NCH
BT0006	Advise Command	N/A	N/T
BT0008	Maintain message files	N/A	N/T QL4 SDP
BT0009	Prepare EMCON plans	N/A	N/T
BT0010	Monitor EMCON	N/A	N/T QL 4 SDP
BT0011	Prepare EMCON Table	N/A	N/T QL 4 SDP
BT0012	Maintain stateboards	N/A	N/T QL 4 SDP
BT0014	Gather Intelligence	N/A	N/T QL 4 SDP
BT0015	Evaluate intelligence information	N/A	N/T QL4 SDP
BT0016	Disseminate intelligence information	N/A	N/T QL4 SDP
BT0017	Record intelligence information	N/A	N/T QL3 SDP
BT0018	Coordinate the display of intelligence information	N/A	N/T QL6B
BT0019	Conduct still photography	N/A	N/T QL3
BT0020	Operate Digital Video Equipment	N/A	N/T QL3
BT0021	Configure Standard Shipboard Display for EW/AWW operations	002	Train
BT0022	Operate SSD for EW/AWW Operations	002	Train
BT0023	Interpret situation display information	002	Train
BT0024	Detect radar contacts	N/A	N/T QL5B
BT0025	Classify radar contacts	N/A	N/T QL5B
BT0026	Input track info into CCS	N/A	N/T QL5B
BT0029	Identify equipment fault/casualty	N/A	N/T QL4
BT0030	Resolve equipment fault/casualty	N/A	N/T QL4
BT0031	Execute equipment prioritization during power failures	N/A	N/T QL5B
BT0032	Maintain physical security of restricted access areas	N/A	N/T QL3

		PO	QL5 Substantiation
BT0033	Maintain a confidential standard document (CSD) loan holder account	N/A	N/T QL5B
BT0034	Conduct muster of classified publications, material and equipment	N/A	N/T QL3
BT0035	Page check publications	N/A	N/T QL3
BT0036	Change/amend publications	N/A	N/T QL3
BT0037	Dispose of classified material	N/A	N/T QL3
BT0038	Perform pre-sail checks on equipment	004, 005	Train
BT0039	Locate and extract information from classified sources	N/A	N/T QL4 SDP
BT0040	Direct the Sensor Section in intelligence gathering	N/A	N/T QL6A
BT0041	Generate Operational messages	001	Train
BT0042	Draft Operational messages	001	Train
BT0044	Extract information from Operational messages	N/A	N/T QL4 SDP
BT0045	Utilize message drafting software and utilities	001	Train
BT0046	Advise Combat Team	N/A	N/T QL4 SDP
CT0001	Flash up/shut down Communication Intercept equipment	N/A	N/T NCH
CT0002	Conduct Communication Intercept equipment functional checks	N/A	N/T NCH
CT0003	Operate Communication Intercept equipment	N/A	N/T NCH
CT0004	Flash up/shut down Electronic Support (ES) equipment	N/A	N/T NCH
CT0005	Conduct ES equipment operator functional checks	N/A	N/T NCH
CT0006	Operate ES equipment	N/A	N/T NCH
CT0007	Edit ES library tapes	003	Train
CT0008	Flash up/shut down Electronic Intercept (ELINT) equipment	N/A	N/T NCH
CT0009	Conduct ELINT equipment operator functional checks	N/A	N/T NCH
CT0010	Operate ELINT equipment	N/A	N/T NCH
CT0011	Edit ELINT library	N/A	N/T QL5B
CT0012	Record electronic emissions	003	Train
CT0013	Track electronic intercepts	N/A	N/T NCH
CT0014	Extract parametric data from databases	N/A	N/T NCH
CT0015	Identify electronic intercepts	N/A	N/T QL4 SDP
CT0016	Classify electronic intercepts	N/A	N/T QL4 SDP
CT0017	Fingerprint electronic emissions	N/A	N/T QL4 SDP
CT0018	Operate PC based EW software applications	N/A	N/T NCH
CT0019	Operate database applications for analysis purposes	N/A	N/T QL4 SDP
CT0020	Analyzing electronic emissions	003	Train
CT0021	Maintain EW tactical logs	N/A	N/T NCH
CT0022	Report electronic intercepts	N/A	N/T NCH
CT0023	Associate electronic intercept information	N/A	N/T QL4 SDP
CT0024	Conduct hull to emitter correlation (HULTEC)	N/A	N/T QL4 SDP
CT0025	Complete EW reports	N/A	N/T QL4
CT0026	Compile electronic order of battle (EOB)	N/A	N/T QL4 SDP

		PO	QL5 Substantiation
CT0027	Compile operational order of battle (OOB)	N/A	N/T QL4 SDP
CT0028	Prepare EW brief utilizing presentation application	N/A	N/T QL4 SDP
CT0029	Perform Electronic Attack (EA) decoys loading procedures	N/A	N/T QL4 SDP
CT0030	Direct EA decoy loading procedures	003	Train
CT0031	Conduct EA decoy firings as Supervisor	N/A	N/T
CT0032	Flash up/shut down Electronic Attack (EA) Systems	003	Train
CT0033	Conduct EA systems operational functional checks	003	Train
CT0034	Operate EA systems	003	Train
CT0036	Coordinate ship ES, EA and EP operations	N/A	N/T
CT0037	Coordinate local EW operations	N/A	N/T
CT0038	Brief the AWW team on EW tactical procedures	N/A	N/T
CT0039	Manage ELINT Files	003	Train
CT0040	Manage Communication Intercept System Networks	N/A	N/T
CT0041	Direct ship ES, EA and EP operations	N/A	N/T
CT0043	Carryout EP measures	003	Train
CT0044	Apply EMCON Rules of Engagement (ROE) policies	N/A	N/T
CT0045	Apply promulgated tactical information as it applies to internal and external organizations	003	Train
CT0046	Approximate the physical characteristics and capabilities of threat systems	003	Train
CT0047	Assessing jamming	N/A	N/T
CT0048	Calculate jamming formula	003	Train
CT0049	Calculating parametric values	003	Train
CT0050	Compile electronic intelligence (ELINT) reports	003	Train
CT0051	Conduct briefings on advanced ELINT	003	Train
CT0052	Conduct EW operational training	N/A	N/T
CT0053	Conduct in-depth electronic pulse train analysis	003	Train
CT0054	Conduct pulse train analysis to third line level	003	Train
CT0055	Correlate analyzed data	003	Train
CT0056	Develop and modify EW tactics	N/A	N/T
CT0057	Disseminate ELINT information	003	Train
CT0058	Execute EW operational training	003	Train
CT0059	Execute Overt/Covert ELINT collection operations	003	Train
CT0060	Execute Overt/Covert EW collection operations	003	Train
CT0061	Execute Overt/Covert Video/Still collection operations	003	Train
CT0062	Identify sources of EW information	003	Train
CT0063	Interpret Electronic Warfare orders	003	Train
CT0064	Interpret Emission Control (EMCON) policies	003	Train
CT0066	Manipulate EW software	003	Train
CT0067	Optimize EW sensor employment	003	Train
CT0068	Perform Area of Interest (AOI) monitoring	003	Train
CT0069	Perform Electronic deception	003	Train
CT0070	Perform radar Pulse Train analysis	003	Train
CT0071	Perform signal analysis	003	Train
CT0072	Perform threat analysis	003	Train

		PO	QL5 Substantiation
CT0073	Program jammer	N/A	N/T
CT0074	React to changes in the EW environment	003	Train
CT0075	Recommend tactical employment of sensors	N/A	N/T
CT0076	Resolve Ambiguities	003	Train
CT0077	Solve problems using mathematical formula and radar parameters	003	Train
CT0078	Track High Interest Vessels	003	Train
CT0079	Utilize Electronic Warfare in support of Blue Water operations	003	Train
CT0080	Utilize Electronic Warfare in support of Land Based operations	003	Train
CT0081	Utilize Electronic Warfare in support of Littoral operations	003	Train
CT0082	Utilize Electronic Warfare in support of Unconventional Warfare and Covert operations	003	Train
CT0083	Utilize search engines	003	N/T
CT0084	Flash up/shut down Electro Optic Detection System	N/A	N/T
CT0085	Configure Electro Optic Detection System	N/A	N/T
CT0086	Operate Electro Optic Detection System	N/A	N/T
CT0087	Resolve Electro Optic System faults	N/A	N/T
CT0088	Flash up/shut down Infrared Detection System	N/A	N/T
CT0089	Configure Infrared Detection System	N/A	N/T
CT0090	Operate Infrared Detection System	N/A	N/T
CT0091	Resolve Infrared Detection System faults	N/A	N/T
DT0001	Flash up/shut down Fire Control (FC) equipment	N/A	N/T QL4 SDP
DT0002	Conduct FC equipment operator functional checks	N/A	N/T QL4 SDP
DT0003	Operate Shipborne Above Water Warfare Weapons and Associated Equipment	N/A	N/T QL4 SDP
DT0004	Collect FC system meteorological information	N/A	N/T QL4 SDP
DT0006	Conduct radar anti-jamming procedures	N/A	N/T QL4 SDP
DT0007	Supervise FC systems operation	N/A	N/T
DT0008	Conduct FC system tracking and assessments	N/A	N/T QL4 SDP
DT0009	Operate weapon systems	004	Train
DT0010	Conduct data logging procedures	N/A	N/T QL4 SDP
DT0011	Collection of post weapon firing data	N/A	N/T QL4 SDP
DT0012	Conduct AA gun firing using primary drill procedures	N/A	N/T
DT0013	Conduct AA gun firing using secondary drill procedures	N/A	N/T
DT0014	Conduct surface gun firing using primary drill procedures	N/A	N/T
DT0015	Conduct surface gun firing using secondary drill procedures	N/A	N/T
DT0016	Conduct Surface to Air missile firing using primary drill procedures	N/A	N/T
DT0017	Conduct Surface to Air missile firing using secondary drill procedures	N/A	N/T
DT0018	Conduct surface to surface missile firing using primary drill procedures	N/A	N/T

		PO	QL5 Substantiation
DT0019	Conduct surface to surfaced missile firing using secondary drill procedures	N/A	N/T
DT0020	Conduct weapon systems misfire procedures	N/A	N/T
DT0021	Prepare surface targets for gunnery exercises	N/A	N/T QL4 SDP
DT0022	Conduct SASS Missile firing using secondary drill procedures	N/A	N/T
DT0023	Conduct SASS Missile firing using primary drill procedures	N/A	N/T
DT0024	Conduct CIWS Surface Engagement	N/A	N/T
DT0025	Conduct CIWS Air Engagement	N/A	N/T
DT0026	Conduct Naval Fire Support	N/A	N/T
DT0027	Operate Naval Fire Support Software	N/A	N/T QL4 SDP
DT0028	Perform AA gun firing using primary drill procedures	N/A	N/T QL4 SDP
DT0029	Perform AA gun firing using secondary drill procedures	N/A	N/T QL4 SDP
DT0030	Perform surface gun firing using primary drill procedures	N/A	N/T QL4 SDP
DT0031	Perform surface gun firing using secondary drill procedures	N/A	N/T QL4 SDP
DT0032	Perform Surface to Air missile firing using primary drill procedures	N/A	N/T QL4 SDP
DT0033	Perform Surface to Air missile firing using secondary drill procedures	N/A	N/T QL4 SDP
DT0036	Perform weapon systems misfire procedures	N/A	N/T QL4 SDP
DT0037	Perform SASS Missile firing	N/A	N/T QL4 SDP
DT0038	Perform SASS Missile firing using primary drill procedures	N/A	N/T QL4 SDP
DT0039	Perform CIWS Surface Engagement	004	Train
DT0040	Perform CIWS Air Engagement	004	Train
DT0041	Perform Naval Fire Support	N/A	N/T NCH
DT0042	Perform FC equipment operator functional checks	N/A	N/T QL4 SDP
DT0043	Perform radar anti-jamming procedures	N/A	N/T QL4 SDP
DT0044	Perform FC system tracking and assessments	N/A	N/T QL4 SDP
DT0045	Perform data logging procedures	N/A	N/T QL4 SDP
DT0046	Flash up/shut down CIWS RCS	004	Train
ET0002	Prepare Above Water Warfare (AWW) exercise support requests	001	Train
ET0003	Prepare AWW firing orders	N/A	N/T
ET0004	Conduct AWW weapon systems pre-firing briefs	N/A	N/T
ET0005	Evaluate AWW team performance	N/A	N/T
ET0006	Provide Tactical recommendations based on Rules of Engagement (ROE)	003	Train
ET0007	Provide Tactical recommendations based on Levels of Force (LOF)	N/A	NT
ET0008	Assess AWW threat information	003	Train
ET0009	Direct the employment of AWW weapons and sensors	N/A	N/T
ET0010	Coordinate compilation of RMP	N/A	N/T
ET0012	Coordinate Local AAW (LAAW)	N/A	N/T

		PO	QL5 Substantiation
ET0013	Coordinate Anti-Surface Warfare (ASuW)	N/A	N/T
ET0014	Coordinate over the horizon targeting (OTHT) engagements	N/A	N/T
ET0015	Tactically employ A/C assigned to force	N/A	N/T
ET0016	Act as Designated Safety Observer (DSO) for weapon firings	N/A	N/T
ET0017	Utilize PC based training systems	N/A	N/T QL3
ET0022	Contribute to databases	N/A	N/T QL3
FT0001	Administer Database(s)	N/A	N/T
FT0002	Advise Database Administrator on Database(s) issues	N/A	N/T
FT0003	Conduct basic ELINT Analysis	N/A	N/T
FT0004	Conduct advanced ELINT Analysis	N/A	N/T
FT0006	Conduct Quality Assurance (QA) on information input in classified database(s)	N/A	N/T
FT0007	Conduct satellite and digital imagery analysis	N/A	N/T
FT0008	Conduct/Participate in technical evaluations, tests and trials	N/A	N/T
FT0009	Conduct/Participate threat analysis including the collection, analysis and dissemination of electronic data	N/A	N/T
FT0010	Flash up Unix system	N/A	N/T
FT0011	Input information into classified database(s)	N/A	N/T
FT0012	Input Software Trouble Reports (STR) on Database(s)	N/A	N/T
FT0013	Input tactical information utilizing GCCS / C2PC	N/A	N/T
FT0014	Operate ELINT equipment for measurement requirements	N/A	N/T
FT0015	Operate EO/IR equipment for measurement requirements	N/A	N/T
FT0016	Operate ESM equipment for measurement requirements	N/A	N/T
FT0017	Operate Unix Systems	N/A	N/T
FT0018	Prepare ELINT equipment for measurement requirements	N/A	N/T
FT0019	Prepare ELINT equipment for operational requirements. (Including Crypto etc)	N/A	N/T
FT0020	Prepare EO/IR equipment for measurement requirements	N/A	N/T
FT0021	Prepare ESM equipment for measurement requirements	N/A	N/T
FT0022	Prepare information to be Input information into classified database(s)	N/A	N/T
FT0023	Process information requested by Request For Information (RFI)	N/A	N/T
FT0024	Process information required by standing requirements	N/A	N/T
FT0025	Process tactical information utilizing GCCS / C2PC	N/A	N/T
FT0026	Produce ELINT reports	N/A	N/T
FT0027	Produce refined imagery reports	N/A	N/T

		PO	QL5 Substantiation
FT0028	Produce reports associated with technical evaluations, tests and trials	N/A	N/T
FT0029	Provide information requested by Request For Information (RFI)	N/A	N/T
FT0030	Provide information required by standing requirements	N/A	N/T
FT0031	Recommend changes to Database(s)	N/A	N/T
FT0032	Represent Canadian Interests in Databases(s)	N/A	N/T
FT0033	Utilize Canadian Forces Electronic Warfare Database (CFEWDB)	N/A	N/T
FT0034	Utilize Canadian Forces Intercept Database (CFIDB)	N/A	N/T
FT0035	Utilize ELINT equipment for operational requirements	N/A	N/T
FT0036	Utilize Maritime Electronic Warfare Library Generation System (MEWLGS)	N/A	N/T
FT0037	Utilize NATO Emitter Database (NEDB)	N/A	N/T
FT0038	Utilize Naval Electronic Warfare Database (NEWDB)	N/A	N/T
FT0039	Utilize Naval Intercept Database (NIDB)	N/A	N/T
FT0040	Verify information input in classified database(s)	N/A	N/T
FT0041	Draft operational correspondence	N/A	N/T
FT0042	Compile intelligence plot	N/A	N/T
FT0043	Operate C4I Inter-Ship circuits	N/A	N/T
FT0044	Develop web pages	N/A	N/T
FT0045	Publish web pages	N/A	N/T
FT0046	Maintain Ship’s/Unit's Classified Websites(s)	N/A	N/T
FT0047	Conduct data transfer via File Transfer Protocol (FTP)	N/A	N/T
FT0049	Develop Local Operating Procedures (LOPs)	N/A	N/T
FT0050	Identify Maritime platforms	N/A	N/T
FT0051	Conduct dissemination of All-Source Information	N/A	N/T
FT0052	Operate GCCS-M	N/A	N/T
FT0053	Operate collaboration at sea tools (to include Chat, GCCS-M Opnotes etc, C2PC)	N/A	N/T
FT0054	Use information from Operational Correspondence	N/A	N/T
FT0055	Process tactical information utilizing C4I systems	N/A	N/T
FT0056	Input tactical information utilizing C4I systems	N/A	N/T
FT0057	Maintain operational logs	N/A	N/T
FT0058	Maintain Recognized Maritime Picture (RMP)	N/A	N/T
FT0059	Operate C4I systems	N/A	N/T
FT0060	Web page design/management	N/A	N/T
FT0061	Conduct CRYPTO changeovers	N/A	N/T
FT0062	Compile SAR information	N/A	N/T
FT0063	Display intelligence information	N/A	N/T
FT0065	Correlate electronic intercept information	N/A	N/T
FT0066	Interpret radar display information	N/A	N/T
FT0067	Amplify track information	N/A	N/T
FT0068	Operate CRYPTO equipment	N/A	N/T
FT0069	Conduct Search and Rescue (SAR)/ Distress/ SUBSMASH communications procedures	N/A	N/T

		PO	QL5 Substantiation
FT0070	Coordinate SAR operations	N/A	N/T
FT0071	Ensure adherence to the procedures of the Canadian Forces Individual Training and Education System (CFITES)	N/A	N/T
FT0072	Create classified and unclassified messages	N/A	N/T
FT0073	Review classified and unclassified messages	N/A	N/T
FT0074	Extract information from classified and unclassified messages	N/A	N/T
FT0075	Maintain message files (electronic/paper)	N/A	N/T
FT0076	Conduct classified employment (complete details of employment are confidential)	N/A	N/T
HT0001	Operate simulator/trainer systems	N/A	N/T
HT0002	Create training scenarios	N/A	N/T
HT0003	Flash up/shut down simulator/trainer systems	N/A	N/T
HT0004	Update training scenarios	N/A	N/T
HT0005	Input training scenarios	N/A	N/T
HT0006	Assist in individual training events	N/A	N/T
HT0007	Assist in team training events	N/A	N/T
HT0008	Conduct individual training events	N/A	N/T
HT0009	Conduct team training events	N/A	N/T
HT0010	Assess individual training events	N/A	N/T
HT0011	Assess team training events	N/A	N/T
HT0012	Evaluate individual training events	N/A	N/T
HT0013	Evaluate team training events	N/A	N/T
HT0014	Correct individual training deficiencies	N/A	N/T
HT0015	Correct team training deficiencies	N/A	N/T
ST0001	Account for classified/designated material	N/A	N/T
ST0002	Act as a mediator	N/A	N/T
ST0003	Action course critiques	N/A	N/T
ST0004	Adhere to environmental protection policies and directives	N/A	N/T
ST0005	Adjust plans, procedures, standards and objectives	N/A	N/T
ST0006	Administer a course critique	N/A	N/T
ST0007	Administer initial counselling (IC); recorded warning (RW) or counselling and probation (C&P)	N/A	N/T
ST0010	Administer/mark tests and examinations	N/A	N/T
ST0012	Amend publications	N/A	N/T
ST0013	Analyze financial expenditures and results	N/A	N/T
ST0014	Analyze planning information	N/A	N/T
ST0015	Analyze unit performance	N/A	N/T
ST0016	Apply IT Security policies	N/A	N/T
ST0017	Approve a course report (CF 377)	N/A	N/T
ST0018	Approve a lesson plan	N/A	N/T
ST0020	Approve a Performance Evaluation Report or a Personnel Development Review	N/A	N/T
ST0021	Approve a Qualification Standard Plan	N/A	N/T
ST0022	Approve a Trg Def	N/A	N/T

		PO	QL5 Substantiation
ST0023	Approve administrative career action for subordinates	N/A	N/T
ST0024	Approve course schedules	N/A	N/T
ST0025	Approve financial expenditures	N/A	N/T
ST0026	Approve goals and objectives	N/A	N/T
ST0027	Approve operational orders	N/A	N/T
ST0028	Approve project schedules/milestones	N/A	N/T
ST0029	Approve training tests / examinations	N/A	N/T
ST0030	Approve unit/section administrative orders	N/A	N/T
ST0031	Assemble planning information	N/A	N/T
ST0032	Assess budget performance	N/A	N/T
ST0033	Assess deportment of subordinates	N/A	N/T
ST0034	Assess dress of subordinates	N/A	N/T
ST0035	Assess effectiveness of training	N/A	N/T
ST0036	Assess effects of changes to financial allocations	N/A	N/T
ST0037	Assess evaluator performance	N/A	N/T
ST0038	Assess impact of changes in resource allocation	N/A	N/T
ST0039	Assess occupational structure/functions	N/A	N/T
ST0040	Assess performance of personnel under stress	N/A	N/T
ST0041	Assess personnel during training activities	N/A	N/T
ST0042	Assess proposed changes to unit Standard Operating Procedures	N/A	N/T
ST0043	Assess security procedures	N/A	N/T
ST0044	Assess suitability of personnel for specific work assignments	N/A	N/T
ST0045	Assess the effects of financial restraint measures	N/A	N/T
ST0046	Assign an instructor	N/A	N/T
ST0047	Assign instructional resource and material (e.g. classroom/equipment/training areas)	N/A	N/T
ST0049	Assign resources to activities	N/A	N/T
ST0050	Assist in the conduct of a summary trial	N/A	N/T
ST0051	Assist in the preparation of work-up programmes	N/A	N/T
ST0052	Assist personnel with a Redress of Grievance	N/A	N/T
ST0053	Authorize personnel qualifications	N/A	N/T
ST0054	Authorize Primary Reserve unit pay sheets	N/A	N/T
ST0055	Brief and debrief personnel with regards to Performance Evaluation Report or Personnel Development Review results	N/A	N/T
ST0056	Command Unit/Department/Section	N/A	N/T
ST0057	Compare results with established objectives	N/A	N/T
ST0058	Compile / Distribute Unit Combat Readiness (CRR) reports	N/A	N/T
ST0059	Compile / Distribute Unit On Job Training (OJT), Ships Development Package(SDP), Pre-Information Package (PIP) reports	N/A	N/T
ST0060	Compile / Distribute Unit Recall Lists	N/A	N/T
ST0061	Complete parade states	N/A	N/T
ST0062	Complete travel documentation	N/A	N/T

		PO	QL5 Substantiation
ST0063	Conduct a Training Review Board (TRB)	N/A	N/T
ST0064	Conduct activities	N/A	N/T
ST0065	Conduct business planning	N/A	N/T
ST0066	Conduct certification boards	N/A	N/T
ST0067	Conduct certification pre-boards	N/A	N/T
ST0068	Conduct classroom instruction/training	N/A	N/T
ST0069	Conduct continuation / proficiency training	N/A	N/T
ST0070	Conduct cost-benefit analyses	N/A	N/T
ST0071	Conduct disciplinary investigations	N/A	N/T
ST0072	Conduct formal inspections of facilities, equipment or work areas	N/A	N/T
ST0073	Conduct general safety checks	N/A	N/T
ST0074	Conduct HQ staff visits/inspections	N/A	N/T
ST0075	Conduct inspections of personnel and quarters	N/A	N/T
ST0076	Conduct interviews	N/A	N/T
ST0077	Conduct inventory checks	N/A	N/T
ST0078	Conduct local purchase order (LPO) transactions	N/A	N/T
ST0079	Conduct mission briefs / debriefs	N/A	N/T
ST0080	Conduct Unit On Job Training (OJT), Ships Development Package(SDP), Pre-Information Package (PIP)	N/A	N/T
ST0081	Conduct practical training exercises	N/A	N/T
ST0082	Conduct proficiency checks	N/A	N/T
ST0083	Conduct risk analysis/risk assessment	N/A	N/T
ST0084	Conduct Summary Investigations (SIs)	N/A	N/T
ST0086	Confirm all transport arrangements	N/A	N/T
ST0087	Consult career managers on personnel matters	N/A	N/T
ST0089	Consult with management on personnel matters	N/A	N/T
ST0092	Control / monitor unit general military programs (e.g. physical fitness, first aid)	N/A	N/T
ST0093	Control resources	N/A	N/T
ST0094	Coordinate acquisition of training resources	N/A	N/T
ST0095	Coordinate assigned unit tasks	N/A	N/T
ST0096	Coordinate career development requirements	N/A	N/T
ST0097	Coordinate course loading with Career Manager	N/A	N/T
ST0098	Coordinate delivery of unit training with outside agencies	N/A	N/T
ST0099	Coordinate first aid training	N/A	N/T
ST0100	Coordinate Primary Reserve support	N/A	N/T
ST0101	Coordinate requirements for meals	N/A	N/T
ST0102	Coordinate support requirements for plans	N/A	N/T
ST0103	Coordinate the environmental assessment process at all levels	N/A	N/T
ST0104	Coordinate training facilities usage with contractors	N/A	N/T
ST0105	Coordinate Unit Chemical Biological Radiological Nuclear (CBRN) training	N/A	N/T
ST0106	Coordinate unit level Environmental Plans, Programs	N/A	N/T

		PO	QL5 Substantiation
	and SOPs		
ST0107	Coordinate Unit Naval Environmental training	N/A	N/T
ST0108	Coordinate Unit Weapons training	N/A	N/T
ST0109	Coordinate visits to Canadian establishments / organizations (e.g. contractor plants, units)	N/A	N/T
ST0110	Coordinate visits to civilian establishments	N/A	N/T
ST0111	Advise senior management on personnel and training matters	N/A	N/T
ST0111	Counsel personnel on personal matters	N/A	N/T
ST0112	Counsel personnel on work related matters (ie: performance, military standards)	N/A	N/T
ST0113	Counsel subordinates	N/A	N/T
ST0114	Counsel subordinates on professional development	N/A	N/T
ST0115	Debrief personnel under training	N/A	N/T
ST0116	Determine alternative course(s) of action	N/A	N/T
ST0117	Determine effect of additional tasking/duties on operations	N/A	N/T
ST0118	Determine effect of additional taskings/duties on activities	N/A	N/T
ST0119	Determine employment rotation of personnel	N/A	N/T
ST0120	Determine equipment/supply requirements for operations/tasking	N/A	N/T
ST0121	Determine goals and objectives for planning purposes	N/A	N/T
ST0122	Determine impact of constraints on operations	N/A	N/T
ST0123	Determine individual education requirements for personnel	N/A	N/T
ST0124	Determine resource requirements for activities	N/A	N/T
ST0125	Determine resource requirements for exercises /courses	N/A	N/T
ST0126	Determine subordinate's individual training and educational requirement	N/A	N/T
ST0127	Develop a Performance Check (PC) or an Enabling Check (EC)	N/A	N/T
ST0128	Develop a Qualification Standard Plan (QSP) (board member)	N/A	N/T
ST0129	Develop briefs	N/A	N/T
ST0130	Develop course critiques	N/A	N/T
ST0133	Develop Unit On Job Training (OJT), Ships Development Package(SDP), Pre-Information Package (PIP) / Operational Training programs	N/A	N/T
ST0134	Develop plans to improve occupational training	N/A	N/T
ST0135	Develop Standard Operating Procedures (SOPs)	N/A	N/T
ST0136	Develop strategies to implement departmental policies	N/A	N/T
ST0137	Develop student study materials and training aids	N/A	N/T
ST0138	Develop subordinates	N/A	N/T
ST0139	Dispose of / route classified documents	N/A	N/T
ST0140	Distribute information bulletins / notices	N/A	N/T

		PO	QL5 Substantiation
ST0141	Draft a course report (CF 377)	N/A	N/T
ST0142	Draft a lesson plan	N/A	N/T
ST0143	Draft a Personnel Evaluation Report (PER) or a Personnel Development Review (PDR)	N/A	N/T
ST0144	Draft a Qualification Standard Plan	N/A	N/T
ST0145	Draft a Training Deficiency (Trg Def)/Trainee Experiencing Difficulty (TED) report	N/A	N/T
ST0146	Draft Record of Disciplinary Proceedings (RDP) (e.g. charge reports)	N/A	N/T
ST0147	Enforce Automated Data Processing (ADP) security procedures	N/A	N/T
ST0148	Enforce deportment of subordinates	N/A	N/T
ST0149	Enforce discipline, deportment and dress regulations	N/A	N/T
ST0150	Enforce general safety programs	N/A	N/T
ST0151	Enforce provisions of access to information and privacy acts (e.g. Release of Fire reports)	N/A	N/T
ST0152	Enforce the application of the Naval Divisional System	N/A	N/T
ST0155	Ensure staff qualifications are maintained	N/A	N/T
ST0156	Ensure standards are maintained	N/A	N/T
ST0157	Establish a training priority	N/A	N/T
ST0158	Establish activity priorities	N/A	N/T
ST0159	Establish budget priorities	N/A	N/T
ST0160	Establish directives and procedures for unit activities	N/A	N/T
ST0161	Establish evaluation standards	N/A	N/T
ST0162	Establish goals and objectives for personnel	N/A	N/T
ST0163	Establish human resource priorities and requirements	N/A	N/T
ST0164	Establish implementation schedules and target dates	N/A	N/T
ST0165	Establish performance standards	N/A	N/T
ST0166	Establish the ranking/selection criteria of personnel for a merit board	N/A	N/T
ST0167	Establish unit manning priorities	N/A	N/T
ST0168	Establish/maintain inventory levels of parts / equipment	N/A	N/T
ST0169	Estimate resource requirements and costs	N/A	N/T
ST0170	Evaluate a course critique	N/A	N/T
ST0171	Evaluate instructor performance	N/A	N/T
ST0173	Evaluate reports to determine action required	N/A	N/T
ST0174	Evaluate standards/criteria used in the workplace	N/A	N/T
ST0175	Evaluate student performance	N/A	N/T
ST0176	Evaluate unit training requirements	N/A	N/T
ST0177	Evaluate workplace for compliance with policies and regulations	N/A	N/T
ST0178	Execute activities	N/A	N/T
ST0179	Explain goals and objectives to subordinates	N/A	N/T
ST0180	Facilitate syndicate exercises	N/A	N/T
ST0181	Forecast manpower availability	N/A	N/T
ST0182	Forecast resource requirements for activities	N/A	N/T

		PO	QL5 Substantiation
ST0183	Forecast transportation requirements	N/A	N/T
ST0184	Formulate unit training and employment policies	N/A	N/T
ST0185	Give briefings/debriefings and speeches	N/A	N/T
ST0186	Identify bilingual position requirements	N/A	N/T
ST0187	Identify personnel qualification requirements	N/A	N/T
ST0189	Identify production problems/backlogs	N/A	N/T
ST0190	Identify reasons for failure to meet objectives	N/A	N/T
ST0191	Implement administrative policies, directives or procedures	N/A	N/T
ST0193	Implement security Standard Operating Procedures	N/A	N/T
ST0194	Inform subordinates of legal rights under military justice system	N/A	N/T
ST0195	Initiate activities to enhance morale	N/A	N/T
ST0197	Initiate administrative action (Initial Counselling (IC); Recorded Warning (RW) or Counselling and Probation (C&P)	N/A	N/T
ST0199	Initiate administrative career action for subordinates	N/A	N/T
ST0200	Initiate corrective action during individual training	N/A	N/T
ST0201	Inspect work areas for security and cleanliness	N/A	N/T
ST0202	Instruct personnel (e.g. formal coursing, On Job Training (OJT), Ships Development Package(SDP), Pre-Information Package (PIP) etc.)	N/A	N/T
ST0203	Investigate a formal or informal complaint (e.g. harassment, disciplinary)	N/A	N/T
ST0204	Investigate accidents and safety violations	N/A	N/T
ST0205	Investigate operating procedural errors	N/A	N/T
ST0207	Invigilate a Performance Check or an Enabling Check	N/A	N/T
ST0208	Issue orders	N/A	N/T
ST0209	Liaise with career managers on personnel matters	N/A	N/T
ST0210	Maintain correspondence files	N/A	N/T
ST0211	Maintain Divisional notes and Unit Employment Record (UER)	N/A	N/T
ST0212	Maintain unit or section reference library	N/A	N/T
ST0214	Maintain unit training records	N/A	N/T
ST0215	Manage an electronic data base	N/A	N/T
ST0217	Manage facilities (e.g. buildings, infrastructures etc.)	N/A	N/T
ST0218	Manage instructors	N/A	N/T
ST0219	Manage NCM professional development	N/A	N/T
ST0220	Manipulate Databases	N/A	N/T
ST0221	Monitor an instructor	N/A	N/T
ST0222	Monitor environmental and energy conservation measures	N/A	N/T
ST0223	Monitor establishment strength	N/A	N/T
ST0224	Monitor expenditures	N/A	N/T
ST0225	Monitor instructors	N/A	N/T
ST0226	Monitor NCM production	N/A	N/T
ST0228	Monitor Performance Evaluation Reports (PERs)	N/A	N/T
ST0229	Monitor Qualification Standard Plan compliance	N/A	N/T

		PO	QL5 Substantiation
ST0230	Monitor the occupation Preferred Manning Level (PML) (e.g. recruitment, promotion, attrition)	N/A	N/T
ST0232	Monitor the training progress of personnel	N/A	N/T
ST0233	Monitor training throughput	N/A	N/T
ST0234	Monitor work-up evolutions	N/A	N/T
ST0235	Navigate on the internet	N/A	N/T
ST0236	Nominate personnel for courses	N/A	N/T
ST0237	Operate Computer applications software	N/A	N/T
ST0239	Organize meetings, conferences or seminars	N/A	N/T
ST0241	Organize summary trials	N/A	N/T
ST0242	Organize unit work relationships and reporting lines	N/A	N/T
ST0243	Participate as a member of a certification board	N/A	N/T
ST0244	Participate as a member of a certification pre-board	N/A	N/T
ST0245	Participate as a member of a Training Plan Writing Board (TPWB)	N/A	N/T
ST0246	Participate in mission brief / debrief	N/A	N/T
ST0247	Participate in parades and military ceremonies	N/A	N/T
ST0248	Participate in physical fitness activities	N/A	N/T
ST0249	Participate in progress review boards	N/A	N/T
ST0250	Participate in recruiting activities (e.g. Recruiter/ Military Career Counsellor/Public Relations Representative)	N/A	N/T
ST0251	Participate in Summary Trials	N/A	N/T
ST0252	Participate in Uunit / section merit ranking of personnel	N/A	N/T
ST0253	Perform as a member of the base duty watch organization	N/A	N/T
ST0254	Perform the duties of a member of the mess committee	N/A	N/T
ST0257	Perform the duties of the President of the Mess Committee (PMC)	N/A	N/T
ST0258	Plan / design training exercises	N/A	N/T
ST0259	Plan activities	N/A	N/T
ST0260	Post or attach post personnel	N/A	N/T
ST0261	Prepare administrative briefings	N/A	N/T
ST0263	Prepare awards and honours submissions	N/A	N/T
ST0264	Prepare briefings/debriefings and speeches	N/A	N/T
ST0265	Prepare correspondence and documents	N/A	N/T
ST0266	Prepare course joining instructions	N/A	N/T
ST0267	Prepare course reports	N/A	N/T
ST0268	Prepare course schedules	N/A	N/T
ST0269	Prepare decision / information briefing notes	N/A	N/T
ST0270	Prepare documentation for career action on military personnel (e.g. re-engagement, LOTP)	N/A	N/T
ST0275	Prepare duty rosters	N/A	N/T
ST0276	Prepare equipment for shipment or storage	N/A	N/T
ST0277	Prepare Establishment Change Proposals (ECPs)	N/A	N/T
ST0278	Prepare Establishment Changes (ECs)	N/A	N/T

		PO	QL5 Substantiation
ST0279	Prepare financial forecasts	N/A	N/T
ST0280	Prepare general administrative reports and returns	N/A	N/T
ST0281	Prepare leave plots for unit personnel	N/A	N/T
ST0282	Prepare lesson plans	N/A	N/T
ST0283	Prepare military correspondence	N/A	N/T
ST0284	Prepare military job descriptions / terms of reference	N/A	N/T
ST0285	Prepare minutes of meetings, conferences or seminars	N/A	N/T
ST0286	Prepare operating budgets	N/A	N/T
ST0287	Prepare operational training exercises	N/A	N/T
ST0288	Prepare orders	N/A	N/T
ST0290	Prepare Performance Development Reports (PDRs)	N/A	N/T
ST0291	Prepare Performance Evaluation Reports (PERs)	N/A	N/T
ST0292	Prepare personnel posting/employment plots	N/A	N/T
ST0293	Prepare Post-Action report	N/A	N/T
ST0294	Prepare Reports and Returns	N/A	N/T
ST0295	Prepare Special parties board	N/A	N/T
ST0296	Prepare Statement of Operational Capability Deficiencies (SOCD)	N/A	N/T
ST0297	Prepare statement of qualifications for positions	N/A	N/T
ST0299	Prepare training tests / examinations	N/A	N/T
ST0300	Prepare unit training plan	N/A	N/T
ST0301	Prepare Unsatisfactory Condition Reports (UCRs)	N/A	N/T
ST0302	Prepare watch and station bill	N/A	N/T
ST0303	Prepare work orders	N/A	N/T
ST0304	Prepare/compile training statistics	N/A	N/T
ST0305	Present administrative briefings	N/A	N/T
ST0306	Process training requests	N/A	N/T
ST0307	Produce a Command annual training plan (Gold Book)	N/A	N/T
ST0308	Produce a departmental annual training plan	N/A	N/T
ST0309	Produce an unit annual training plan	N/A	N/T
ST0310	Project budget preparation / procedures	N/A	N/T
ST0311	Provide advice in matters of supply, administration and training	N/A	N/T
ST0312	Provide advice on manning issues to commanders	N/A	N/T
ST0313	Provide advice to subordinates on administrative and disciplinary matters	N/A	N/T
ST0314	Provide assistance in establishing realistic expectations with subordinates	N/A	N/T
ST0315	Provide input for Command reports and returns	N/A	N/T
ST0316	Provide input to Business Plans (BPs)	N/A	N/T
ST0317	Provide input to grievance hearings	N/A	N/T
ST0318	Provide planning information	N/A	N/T
ST0319	Provide recommendation to Career Review Boards (CRBs)	N/A	N/T
ST0320	Provide recommendation to Training Review Boards	N/A	N/T

		PO	QL5 Substantiation
	(TRBs)		
ST0321	Provide section/unit with feedback	N/A	N/T
ST0322	Provide status of training	N/A	N/T
ST0323	Provide subordinates with feedback	N/A	N/T
ST0324	Receive invoices for payment	N/A	N/T
ST0325	Recommend a disciplinary or an administrative measure in response to a complaint	N/A	N/T
ST0326	Recommend administrative career restrictions for subordinates	N/A	N/T
ST0327	Recommend career options	N/A	N/T
ST0328	Recommend changes to schematics/technical drawings	N/A	N/T
ST0329	Recommend Material Authorization Request (MAR) for approval	N/A	N/T
ST0330	Recommend personnel for Canadian Forces honours, awards and recognition programs	N/A	N/T
ST0331	Recommend personnel for career and / or educational programs	N/A	N/T
ST0332	Recommend personnel for remedial physical training	N/A	N/T
ST0333	Recommend student re-coursing	N/A	N/T
ST0334	Recommend unsatisfactory condition report (UCR) for approval	N/A	N/T
ST0335	Refer media to appropriate agencies	N/A	N/T
ST0336	Refer personnel to professional counselling/social services	N/A	N/T
ST0337	Resolve a formal or informal complaint	N/A	N/T
ST0338	Review / approve course reports	N/A	N/T
ST0339	Review / Recommend Establishment Changes (ECs)	N/A	N/T
ST0340	Review a course report (CF 377)	N/A	N/T
ST0341	Review a lesson plan	N/A	N/T
ST0342	Review administrative orders, instructions or directives	N/A	N/T
ST0343	Review and approve a Training Deficiency	N/A	N/T
ST0344	Review and approve an Establishment Change Proposal (ECP)	N/A	N/T
ST0345	Review briefings and speeches	N/A	N/T
ST0346	Review Command reports and returns	N/A	N/T
ST0347	Review correspondence and documents	N/A	N/T
ST0348	Review Course Training Plans (CTPs)	N/A	N/T
ST0349	Review decision / information briefing notes	N/A	N/T
ST0350	Review general administrative reports and returns	N/A	N/T
ST0351	Review goals and objectives of plans	N/A	N/T
ST0352	Review Local Operating Orders (LOPs)	N/A	N/T
ST0353	Review military correspondence	N/A	N/T
ST0354	Review military police reports	N/A	N/T
ST0355	Review Occupational Specifications (OS) or Occupational Specialty Specifications (SS)	N/A	N/T
ST0356	Review performance development reports (PDR)	N/A	N/T

		PO	QL5 Substantiation
ST0357	Review Performance Evaluation Reports (PERs)	N/A	N/T
ST0358	Review personnel records and files	N/A	N/T
ST0360	Review recommendations for awards and honours	N/A	N/T
ST0361	Review Standard Operating Procedures (SOPs)	N/A	N/T
ST0362	Review technical publications	N/A	N/T
ST0363	Review Unit SOPs	N/A	N/T
ST0364	Review unit/section administrative orders	N/A	N/T
ST0365	Review/approve general administrative reports and returns	N/A	N/T
ST0367	Review/edit correspondence and documents	N/A	N/T
ST0369	Perform duties of Assisting Officer	N/A	N/T
ST0370	Schedule personnel for duty	N/A	N/T
ST0371	Secure classified/designated documents or equipment	N/A	N/T
ST0372	Select personnel for courses and / or training	N/A	N/T
ST0373	Set agenda for meetings/committees/conferences	N/A	N/T
ST0374	Set up course schedules	N/A	N/T
ST0382	Track collective and individual training	N/A	N/T
ST0383	Update nominal rolls	N/A	N/T
ST0384	Update personnel records and files	N/A	N/T
ST0385	Update Unit Employment Records (UERs)	N/A	N/T
ST0386	Upgrade course content	N/A	N/T
ST0387	Utilize computer software	N/A	N/T
ST0388	Validate training programs	N/A	N/T
ST0389	Verify adherence to performance standards	N/A	N/T
ST0390	Visit subordinate units/sections	N/A	N/T
ST0391	Write civilian Performance Review Report (PRR)	N/A	N/T
ST0392	Write military letters	N/A	N/T
ST0393	Conduct Quality Assurance (QA) of Qualification Standard Plans (QSP)	N/A	N/T
ST0394	Conduct Quality Control (QC) of Qualification Standard Plans (QSP)	N/A	N/T
ST0395	Conduct curriculum review	N/A	N/T
ST0396	Create tests and examinations	N/A	N/T
ST0397	Act as role model for subordinates	N/A	N/T
ST0398	Act as role model for students	N/A	N/T
ST0399	Monitor and assess harbour training exercises	N/A	N/T
ST0400	Monitor and assess at sea training exercises	N/A	N/T
ST0401	Conduct duties of Mess Attendant	N/A	N/T
ST0402	Perform On Job Training (OJT) or Ships Development Package(SDP) or Pre-Information Package (PIP)	N/A	N/T

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates:

26 May -27 Jun 2008

Location:

CFFS Esquimalt.

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Zubrinich (TDO)	AGS	Lt (N)	00211	CFFSE
Horner	CP	CPO1	00114	DMAR PERS
Gracey (Chair)	SR	CPO2	00115	DPGR 3-2
McKenzie	DG	CPO2	00299	DMTE 2-2-3
Frerichs (Stds)	WW	PO1	00115	CFFSE
Randell	EB	PO2	00115	CFNOS
Pearce	RW	PO2	00115	CFFSE
Deschenes	E	PO2	00115	HMCS ALGONQUIN
Theriault	SP	MS	00115	CFNOS
Heslop (Sec)	RR	MS	00115	HMCS VANCOUVER

Reference: MSG DTG 221806Z APR 08 (DMTE). Subj: Convening Order - NESOP QL4/5A Qualification Standard And Plan Writing Board (QSPWB) 26 May - 27 Jun 08 CFFSE Esquimalt
Note: Given the dual nature of this QSPWB, this ROP will be used for both the QL4 and NESOP RQ-S1 QSPs. The ROP contain details pertinent to both and should be read prior to the QL3 and QL5BA writing boards scheduled in late ‘08 and the winter of ‘09. The board was convened at CFFSE although CFNOS still retains CCA over this training. Where applicable, CFNOS Standing Orders were used.

Board Objective:

1. The mandate of the board is to review and update specialty specifications (SS) AIXA and AIXB and QSP as required

Specification Task – No Train Decisions:

2. The tasks selected for training are detailed at Annex B of both QL4 and NESOP RQ-S1 QSPs.

Discussions:

Day 1, Monday, 26, May, 2008

3. CPO1 Horner, began with opening remarks. He spoke of the outline of how board will function. CPO2 Gracey, identified the breakdown of how the trade is currently laid out. He specified by each level how the board is going to incorporate new degrees of training. It will include the following breakdown: NES Junior Operator Basic (QL3), NES Junior Operator (QL4), NES Intermediate Operator (QL4 –NESOP RQ-S1), NES Senior Operator (QL5B – QL 6A), NES Supervisor (QL6A), and AWWD (QL6B) with a degree of on the job training to be incorporated in between. He discussed how the new training plan will cover the next 5 years as we are not in a position where we know what we are going to be working with post FELEX re-fit. He emphasised the short term plan will bring us to the point where we will train post FELEX re-fit. CPO2 MacKenzie reiterated that this board will drive the plan. He addressed the fact that the agenda must be clear and when he leaves he intends to have a solid plan in place for how we will go forward.

4. The QSPWB TDO, Lt (N) Zubrinich, provided a brief on the schedule, rolls and responsibilities of each member, and an overview of the aim, objectives, rules, conduct and process required for the board and the QSPWB package contents. He then briefed the board on the Canadian Forces Individual Training and Education System. (CFITES), focusing on the analysis and design phases. He briefed how Scalars will be designed and emphasised that the Scalars will be 90% reflective of the training. He reviewed the Assessment Plan and the TSK Proficiency Levels and how they affect the whole process.

5. CPO2 Gracey, briefly went through all the Job Based Specification (JBS) job titles. He opened the online JBS list and gave an overall review of the different positions and their detailed specifications. He then reviewed the position for Intermediate Operator and the board looked at where things currently stand, setting the stage.

6. Lt (N) Zubrinich, went through expectations/parking lot from each member of the board and what they would like see from the board. A list was then created of items for the board to focus on. It brought forth a discussion about how OJT should be reintroduced. It was identified there is a defined gap in between each course where there is no onboard preparation being done to develop personnel to move forward into their next course level. The list that was generated is as follows;

- Accurate Training (terminology) OJT, PIP. .etc.
- Develop clear and detailed QSP
- Guidance for Trg Pos onboard. (QL4 and up) (standardized)
- Bring back OJT/OJE ect., (SDSP, DL, Trainer) Address current deficiencies in between QLs
- “Generic” Training Reflective of the job.
- Gap in trg level between Sea/Shore postings (i.e. math) (QL5 (S1 and up) Ident common training)
- Proficiency level change does not = training

7. Lt (N) Zubrinich briefed the approach towards how the board will achieve its aim. He reviewed the Scalar process and how they will be constructed as a board. He then brought the board through and example of a practice scalar using changing a tire as an example.

8. The board discussed and identified the likelihood of creating equipment famil (EF) and Ship Development Packages (SDP) in pre and post formal QL training to be run by the formation/occupations. Pre and post course packages (equipment famil and SDP to be developed by **formations, occupations**)

9. Under the guidance of Lt (N) Zubrinich and CPO2 Gracey, The board proceeded with the train – no train list for Junior Operator (QL4)

Day 2, Tuesday, 27,May, 2008

10. The board continued with the analysis and design of training for the QL4. It was noted by the board that there is a distinct difference between Fire Control and CIWS training as it pertains to the Train – No Train list.

11. All tasks in duty area “F” (Operational & Analysis Centres) are deemed No Train as the focus of the training is shore based. All tasks in duty are “H” (Trainer/Simulator) were also deemed the same.

12. The board discussed duty area “D” DT0040 and determined that CIWS training will not be covered until the NESOP RQ-S1 course.

13. The board completed the Train – No Train list and CPO 1 Horner thanked the board for the successful completion of the list.

14. The board proceeded with the QL 4 scalar process and identified the jobs associated with the NESOP Junior Operator;

- Operate Fire Control Systems
- Operate Electro Optic/Infrared Systems (trained based on procurement of EO/IR system)
- Operate Electronic Warfare Intercept Equipment (trained at QL 3)
- Operate Communication Intercept Equipment (trained at QL 3)
- Operate ELINT equipment (trained at QL 3)
- Load CHAFF & Decoy (trained at QL 3)
- Prepare Electronic Order of Battle and Operational Order of Battle (trained at QL 3)

15. The board agreed that the focus of the QL 4 Junior Operator is to operate Fire Control Systems and EO/IR, with the remainder of the QSP either being touched in QL 4 training or more applicable at the QL 3 level. In addition the operator was seen to have a requirement to operate SSD for AWW operations.

16. The board determined that the task CT0039 (Manage ELINT Files) was not applicable at he QL 4 level, as the junior operator does not manage files but instead contributes to data bases. (ET0022).

17. The board completed the scalar process and identified the performance objectives for the QL 4 course:

a. PO 001 Operate SSD for AWW operations

b. PO 002 Operate Fire Control Systems

c. PO 003 Operate EO/IR Systems

18. The board proceeded with the construction of Performance Objectives for the QL 4 course

Day 3, Wednesday, 28, May, 2008

19. The board revisited the introduction of the SDP to address the gap expected / identified between QL's in NES Op career progression. It was identified that the SDP's need to be developed as soon as possible after the QL4 and 5A QSPWBs to provide timely recommendations/corrections to all levels of training. The board also identified that there is an inherent risk associated with not having these SDPs available to address the gaps in between QLs. Therefore, its expected that the SDPs will be complete/approved NLT Dec 08. The SDP needs to be under the authority of the occupation/formation (not a training establishment as it is not "training"), therefore responsibility for package completion rests with the individual units. The board decided that the proper naming convention for the SDPs should be as follows:

- SDP QL3 = Development package to bridge QL3 to QL 4
- SDP QL4 = Development package to bridge QL4 to NESOP RQ-S1

NOTE: See Week 4 comments. The SDP was identified as <u>actual training</u> due to increase in proficiency levels particularly in PO 002 (Operated FC Systems for AWW Operations), vice a "skills development" package. Therefore, two OJPR packages in place of the SDP area to be developed to augment the respective QSP's and QL Training. Respectively, the board will only develop the QL4 OJPR at this time while the QL3 OJPR will be developed during the QL3 Board tentatively set for Nov 08. Having an OJPR will solve curriculum control issues, maintain standardization in the fleet, and close the gaps between courses by linking On-the-job training to existing POs in formal courses.

20. The board finalized writing the POs then continued with completing the motherhood information contained in front of the QSP as well as chapter 1 detail as applicable. It was observed by the board that the resources and course duration calculation will need to be revisited after the pilot course is run and amended as necessary.

21. The TDO then briefed the board on the design phase for the QL 4 training, beginning with discussing the instruction analysis and student assessment details. The day was concluded with a thorough review of all policies (DAODs, MARCORD, NTI and CFNOS SOs) required to accurately identify training assessment and other chap. 3 details.

Day 4, Thursday, 29 May, 2008

22. The board continued the design phase with the scalar process of breaking down the QL 4 course scalar POs into EOs and teaching points using the instructional analysis.

23. The introduction of Math training in the NESOP trade was initially discussed by the senior members of the occupation during Fleet Week 08 CPO1/CPO2 meetings. The overarching requirement for the math packages is based on the implementation and tasks of forthcoming EW and ELINT equipment. CPO2 Gracey advised the board that math had to be introduced into the QL 3/4 courses based on the recommendations of the above mentioned meetings. This prompted the board to discuss how the math training should be put into the course layout. After a lengthy discussion Lt (N) Zubrinich addressed the board with a breakdown of the current 1030 math package, which the board then decided was needed to be taught. The board determined that because the standing 1030 math package is broken into 13 modules, the best way to go about addressing the math issues was to divide the 1030 package between the QL3 and QL4 as follows:

QL 3	QL 4
Mod:	
1) Review Arithmetic and Signed Numbers	7) Solving Equations
2) Powers and Roots	8)Triangle Trigonometry
3) Scientific Notation	9) Vectors
4) Unit Conversion	10) Circular Functions
5) Graphs	11) Logarithms
6) Introduction to Algebra	12) Decibels
	13) Number Systems

24. It is recommended for a QL 3 that math be assessed as a critical EC or somehow incorporated in a PO and subsequent PC IAW MARCORD 9-60. it was discussed that once the QL 4 course was finished there would be a requirement for verifying this knowledge through the successful completion of the 1030 math exam prior to coursing of NESOP RQ-S1. After further discussion the board decided that the math package wasn't directly related to any existing POs therefore decided to increase the level expected in ELINT analysis (task CT0020) at the QL 4 level and create a separate PO titled Analyze Electronic Intercepts.

25. While conducting the analysis of PO 002, Operate FC systems, the board discussed and compared how this material is taught currently, how it is tested, and if this adequately, or over prepares the member for the job they perform at the QL4 level on board. After reviewing the content of PO 002, it was found there are theory and classroom components for both Gun and MSL weapons, plus the Fire control radar / STIR application. Different approaches to how this could best be chunked were discussed, and the board identified two possible courses of action:

- Status Quo: One PO to Teach theory for all STIR, Gun and MSL weaponry, along with applicable written tests, then move on to the practical phase with sufficient practice and developmental before culminating in one summative test : the PC, which combines both Gun and MSL for Air and Surface firing; and

- Option One: Create two POs to teach the general safety theory upfront, but instead of dividing the two sections into Gun and MSL, focus one PO on each

threat (AAW drills and ASuW drills which utilize both Gun and MSL).. Theory

tests could be absorbed into the practical EC's and PC's, with the possible exception of general safety etc. This could better prepare the QL4 earlier to understand the warfare aspect earlier in their career.

- Option Two (a version of Option one): Create one PO with EO's to teach the general safety theory upfront, and the two EOs on each threat (AAW drills and ASuW drills which utilize both Gun and MSL).. Theory tests could be absorbed into the practical Critical EC's and a final PC, similar to the existing scenario. This could better prepare the QL4 earlier to understand the warfare aspect earlier in their career.

- The board decided option two was the preferred course of action.

Day 5, Friday, 30 May 2008

26. The board convened and summarized discussion about how to develop PO 002. The consensus was reached that the best approach for AAW and ASuW would be to split the two and create them as separate ECs.

27. Issues with how to go about teaching safety and whether or not to test it in written form arose. After discussion the board decided that it was best to keep safety as an EC as opposed to making a written PC, as safety will be tested throughout the Practical Phase of testing and can be checked in the form of written quizzes and home work.

28. The board then proceeded with the scalar process and worked through knowledge and tasks to form an effective org chart.

29. The board proceeded with the writing of Chapter 4 lesson specifications starting with PO 001 Operate SSD for AWW Operations.

Day 6, Monday, 2, June, 2008

30. The board continued with creating the lesson specifications for PO 002.

31. The board completed the lesson Specification for PO 002 and continued with PO 003. Some discrepancies were identified with the scalar for PO 002 with regard to how one of the teaching points (BT0030) was worded. After discussion it was decided there would be no changes made.

32. In the afternoon the board continued with PO 003 and PO 004. By end of day PO 004 was completed and a draft of the QSP for the QL 4 Course was completed.

Day 7, Tuesday, 3, June, 2008

33. To accommodate DMTEs requirement to preliminary approve Chapters 1 & 2 for the QL5A, the board continued with the analysis of the NES Intermediate Operator (QL5A) training. The board will revisit the QL4 and QL5A after the lesson specs are completed for both.

34. The board began the scalar process for the analysis phase. There was discussion as to how the trade is to go about Analyzing Electronic Intercepts as we are not currently fitted with front line analysis EW equipment. The TDO advised that we still have to structure the points as it stands right now. The board worked to develop a working scalar.

35. The board began with Chapter 2 of the QSP and began writing the Performance Objectives for the QL 5A course.

36. It was identified by the board that some of the knowledge titles contain the terms ESM, ECM and EPM is old terminology and has been replaced with the terms Electronic Support (ES), Electronic Attack (EA), and Electronic Protect (EP). The term Electronic Warfare (EW) encompasses all three of the terms ES, EA and EP. It is of the board's interest to revisit the titles given for teaching points containing the old terminology ESM, ECM and ECCM.

Day 8, Wednesday, 4, June, 2008

37. CPO2 Gracey had e-mailed both coast Sea Training NESOP CPO2s with regard to whether or not flash up/shutdown should still be included in our training. The board's general consensus was that on ship the CSE dept had taken over flash up/shutdown and it was no longer required to be trained. An e-mail came back from both coast Sea Training NESOP CPO2s and it was stated that there is no set in stone policy as to who is to be in charge of flash up/shutdown. Therefore it will still be included in training at all levels.

38. It was decided by the board that that the headers ES, EA and EP are in fact replacing the older terminology previously discussed on the previous day.

39. The board debated how PO 001 Utilize EW Systems should be laid out. A discussion lead to the debate over how this should taught. In the very near future there will be new EA and EP equipment therefore it was decided that the EOs for PO 001 should reflect ES, EA and EP as headers and the current equipment on ship be trained until which point new equipment comes online.

40. The board discussed the First EO of PO 001, (EO 001.01Operate SSD) and the consensus was that due to the importance of the EO it should be made a PO and over a week of time be dedicated to the teaching and examination of the PO. EO 001.01 has now become PO 002 Operate SSD.

41. The board ended the day with much discussion and discrepancy over how the course should be laid out. One of the outstanding problems is that there is no solid direction on which equipment the Navy will be working with post FELEX. At this point the board determined the JBS required some minor refinement. The board agreed to adjourn and visit everything the next day.

Day 9, Thursday, 5, June, 2008

42. The TDO advised the board that today we need to start fresh and begin by revisiting the JBS and go through, review and amend as necessary the JBS tasks to reflect anticipated jobs of the NESOP. Things he advised us to cover:

- "New" Terminology (EA/EP vs. ECM/ECCM);
- "Verb" selection appropriate to the position;
- 75% current job/equip 25% unknown (future);
- Required vs. need/want; and
- Train as we "fight" 1 in 2 watch (AWW).

43. Under the direction of CPO2 Gracey, the board reviewed the JBS and made appropriate title changes and re-worked the numbering to accommodate the jobs of

NESOPs in the present and future (next 5 years). After the JBS was refined, the board will be able to refocus on the analysis and design of the QL 5A QSP.'

Day 10, Friday, 6, June, 2008

44. CPO 2 Gracey and PO2 Randell started the morning by taking the board through a whole re-organization of the JBS and scalar for the QL 5A. There was a detailed look at the JBS and what tasks and knowledge better tie into the occupation. By the end of the day the board had developed a whole new scalar outline and revisions to the JBS.
45. The changes and amendments made to the JBS were then reviewed by the board and the Train/No Train list was revised.

Day 11, Monday, 9, June, 2008

46. The board reconvened and began on the scalar for the QL5 A. Upon completion the board had developed a course layout differing from the week previous as follows:

- PO 001 Generate Operational Messages
- PO 002 Operate SSD for EW/AWW Operations
- PO 003 Operate EW Equipment
- P0 004 Operate Weapons Systems
- PO 005 Operate Non-RADAR Detection Systems

47. Once the scalar was developed the board continued on with the writing of Chapter 2 of the QL5 A QSP.
48. The Board then reviewed the current acronym lists and reference lists and determined which acronyms and references currently in use will apply to the new QSP.

Day 12, Tuesday, 10, June, 2008

49. The board continued on with the development of Chap. 4 of the QSP while the board secretary prepared the QL 5A Scalar. The day was spent working out the timings and test methods for each of the PO and EOs. The board finished the day with more than half of Chapter 4 written.

Day 13, Wednesday, 11, June, 2008

50. The TDO reviewed our progress on the QSP and went through and provided the board with guidance on some of the fine tuning of the work on Chapter 4 that had been completed so far.
51. The issue of CIWS weapons system certification arose. Direction provided in MARCORD Vol 3, 46-3, table 2-2 states that the FCO is required to be weapons certified. This training can be provided at CFNOS for the applicable class of ship for NES Ops. The board agreed that Weapons System Certification (PC) should be part of PO 004 (Operate weapon Systems) and needs to be invigilated.

Day 14, Thursday, 12, June, 2008

52. The board started the day with continuation on the QSP Chap 4. (EO 005)

53. The board completed Chapter 4 of the QSP and proceeded with the ANNEX A – Main Reference list. The TDO outlined the priorities for the remainder of the board timeline. The “Parking Lot” of issues that had come up in during discussions was revisited and items that had been visited had been removed and any forthcoming issues were added.

Day 15, Friday, 13, June, 2008

54. The board met at AWWTC to go through and cross reference the publications for the reference list. The list was updated and Chapter 2 and Chapter 4 were amended as required.
55. The board conferred with LCdr. McBain (DMPOR3-3) and discussed employment and the delineation of tasks associated with EO/IR. The board recommended the responsible occupation for SEOSS not be the NESOP occupation.

Day 16, Monday, 16, June, 2008

55. The board continued and reviewed chapter 4 of the QL5 A QSP with the sequencing, the allotted timings for all of the EOs as well as costing, resources and acronyms.
56. The board determined how the course would weigh out percentage wise. PO 003 Operate EW Equipment and PO 004 Operate Weapons Systems were seen as the pillars to he QL5 A, Thus given the most weight. When PO 005 Operate Non-Radar Detection Systems is more established the weighting should be revisited to reflect the importance of this subject. The course weighting looks as follows:

PO#	Name	Ranking	Weight (%)
001	Generate Operational Messages	3	10
002	Operate SSD for EW/AWW Operations	3	10
003	Operate EW Equipment	1	40
004	Operate Weapons System	2	35
005	Operate Non-Radar Detection Systems	4	5

57. The board wrapped up the QL 5A by reviewing the document and looking at the various instructional strategies applicable to the QL 5A student. The board determined that the majority of the EOs could be taught using interactive lecture/lectures however, advanced instruction techniques such as guided discussion and case studies could be appropriate for certain teaching points and lessons. These were identified in the remarks paragraph for the relevant EOs.

Day 17, Tuesday, 17, June, 2008

58. The board began with an overall review of the QL 5A QSP. After some small corrections and a table for Annex L (Student Write Up) was created, the board completed the QL 5A QSP.
The Annexes were reviewed and inserted into the document including the sequencing and costing which were completed and included in the Annexes. The course duration was calculated and broken down and justified as follows:

	NEW	OLD
Actual Training Days (ATT)	58	41
Training Administrative Time (TAT) (if required)	3.5	1
Base Administrative Time (BAT)	2	1
Training Travel Time (TTT) (if required)	2	0
Physical Training (PT) IAW CDS Direction 2203.4	5	Not accounted for
Course Duration (total calendar days)	69	43

- ATT was increased due to the addition of Operate SSD for AWW Operations, Operate Non Radar Detection Systems and an increase to Analyze Electronic Intercept to include: Math for SIGINT 1030 modules and weapon system certification. Previously tasks associated with these jobs were covered in higher levels of training but are required for occupational employment at the QL 4 level.

- TAT was calculated IAW MARCORD 9-30 thus increased to reflect a maximum of two periods per week.

- BAT was increased for in clearance and out clearance for members on TD as well as TTT.

- PT was calculated to reflect CDS guidance of 3 periods per week.

59. The board revisited the QL 4 scalar. After the changes and additions were made to the JBS there was a requirement to review the analysis phase for QL 4 training. The board re-worked the scalar with the new knowledge and tasks from the revised JBS. Upon completion the board came up with 4 distinct POs.

1) PO 001 Operate SSD for AWW Operations

2) PO 002 Operate Shipborne AWW Weapons and Associated Equipment

3) PO 003 Operate Non-RADAR Detection Systems

4) PO 004 Analyze Electronic Intercept

Day 18, Wednesday, 18, June, 2008

60. The board spent the morning defining the EO titles and finalized where exactly the knowledge and tasks fell under for the purpose of the scalar.

61. The scalar was completed and captured for the QSP Annex. The board proceeded with Chapter 4 of the QSP and determined the weighting of the POs for the course. The board determined how the course would weight percentage wise. PO 002 Operate Shipborne AWW Weapons and Associated Equipment and PO 004 Analyze Electronic Intercept were seen as the pillars to the QL4, thus given the most weight. When PO 003 Operate Non-Radar Detection Systems is more established the weighting should be revisited to reflect the importance of this subject. The course weighting looks as follows:

PO#	Name	Ranking	Weight (%)

01	Operate SSD for AWW Operations	3	5
02	Operate Shipborne AWW Weapons and Associated Equipment	1	50
03	Operate Non-Radar Detection Systems	4	5
04	Analyze Electronic Intercept	2	40

62. The board discussed the SDP and whether this was actual skills development or formal on the job training which indicates an increase in proficiency levels IAW the JBS. The board chair and TDO reviewed the pros and cons of SDP over OJPR and decided that including an OJPR package for both QL 3 and QL 4 was the most effective way to manage, administer and maintain quality control by the CCA.

Day 19, Thursday, 19, June, 2008

62. The board began the day by finishing Chapter 4 of the QSP and Annex K.
63. The board finished Chapter 4 and Chapter 2.

Day 20, Friday, 20, June, 2008

64. The board started the morning by reviewing the QSP document threading everything together.
65. The Annexes were reviewed and inserted into the document including the sequencing and costing which were completed and included in the Annexes. The course duration was calculated and broken down and justified as follows:

	NEW	OLD
Actual Training Days (ATT)	77	66
Training Administrative Time (TAT) (if required)	4.4	Not accounted for
Base Administrative Time	2	1
PT (IAW CDS Direction 2203.4)	6.6	Not accounted for
Training Travel Time (if required)	2	Not accounted for
Course Duration (total calendar days)	90	67

- ATT was increased by 11 days to accommodate Operate Non Radar Detection Systems, Operate Weapons Systems (CIWS) including Weapon System Certification, and an increase in Operate EW equipment. Previously tasks associated with these jobs were covered in higher levels of training but are required for occupational employment at the QL 5A level.

- TAT was calculated IAW MARCORD 9-30 thus increased to reflect a maximum of two periods per week.

- BAT was increased for in clearance and out clearance for members on TD as well as TTT.

- PT was calculated to reflect CDS guidance of 3 periods per week.

66. The board finished the day reviewing the train/no train list to get started on the construction of the OJPR.

Day 21, Monday, 23, June, 2008

67. The board went through all of the knowledge and tasks from the train/no train list and established the working scalar for the OJPR.

68. Once the scalar was completed the board generated the OJPR Document.

69. The OJPR document was completed (Draft).

Day 22, Tuesday, 24, June, 2008

70. The board reviewed the OJPR standard checklists and it was identified that there needs to be more detail covered in the checklists. It was agreed by the board that in order to cover the topics so that it validates the required standard to meet there needs to be more detail added.

71. After discussing the administrative process of the OJPR once complete, the board agreed to the following:

a. OJPR Package is signed off by the CO;

b. Letter drafted to DMTE, CCA, and DMILC2 that member has finished OJT

c. CFNOS and CFFSE Standards are to invigilate a practical PC as soon as possible after OJT completion, and is to be granted AIXA and MITE ID (TBD) for OJT if PC is passed. (Note 1)

d. Standards to invigilate practical PC on board or alongside (NCOT) (Note 2);

e. One (1) year from formal course (MITE ID 114958) completion, and successful completion of OJPR package (MITE ID TBD), member will be eligible to be course loaded on QL5A.

Note 1: The board discussed at length the impact of a practical PC on fleet resources career progression, and decided it is the better choice to maintain standards and provide the Fleet with a competent operator at the QL4 Level in preparation for their next course. The board voted on whether a practical PC was a valid and justifiable course of action for NES training. Results were 3 (PO1 Frerichs, PO2 Decshenes, MS Theriault) against and 2 (PO2 Randell, PO2 Pearce) for a PC. After further discussion, the Board Chair over ruled the vote based on there being no external check and balance of the unit (CCA needs to retain Quality control, and this can best be achieved through the invigilation of a practical PC, similar to in-house courses.)
Note 2: The board identified that the practical PC will need to be very detailed in checklist development, timings, conditions etc. in order to be effective and a true

assessment of ability. The PC should focus on PO 002 (Operate Shipborne AWW Weapons and Associated Equipment) as this PO reflects the change in proficiency level from 2 to 3 making them an independent operator.

72. The board left the OJPR as a living document and it will be further amended and reviewed when the document is brought to the fleet.

Post QSPWB note:
The decision to develop OJPR as a training approach was overturned in place of a Ships Development package to be administered/under the control of the MOC.

Guidance to Instructional Designers:

SER	ITEM	OPI	Estimated Due Date
QL4 (Pilot Start 5 Jan 09)			
1.	Ensure instruction staff are prepared in Math, Non Radar Detection systems	CFNOS Senior Instructor	NLT End Sep
2.	Develop all EC, Critical EC and PC checklists	CCA Instruction and Standards	15 Nov 08
3.	Finalize EC,Critical EC and PC Checklists	Staff CCA Instruction and Standards	15 Dec 08
4.	Develop Teaching points and Verify reference list	Staff CFNOS	End Aug 08
5.	Develop Master Lesson Plans / Presentation	Instruction staff CFNOS	End Nov 08
6.	OJPR Package (Annex C)	Instruction Staff CFNOS	End Nov 08
7.	Develop OJPR PC	Standards CFNOS	End Nov 08
8.	Adjust CP for course duration	Standards CFNOS	Mid Sep 08
9.	Schedule, plan and book Trainer in prep for pilot course	Training Chief CFNOS Instruction staff	NLT End Sep 08
QL5A (Pilot Start 5 Jan 09			
1.	Ensure instruction staff are prepared in Math, Non Radar Detection systems	CFNOS Senior Instructor	NLT End Sep
2.	Develop all EC, Critical EC and PC checklists	5. Develop Master Lesson Plans / Presentation	
3.	Finalize EC, Critical EC and PC Checklists		
4.	Develop Teaching points and Verify reference list		

Approved: 20 Aug 2015
Version No:2.7
MITE Crse ID:120689

CFNOS	15 Nov 08
Instruction and Standards Staff CFNOS	
Instruction and Standards Staff CFNOS	
Instruction staff CFNOS	15 Dec 08
Instruction Staff	
	End Aug 08
	End Sept 08

SER	ITEM	OPI	Estimated Due Date
6.	Adjust CP for course duration	CFNOS Training Chief	Mid Sep 08
7.	Schedule, plan and book Trainer in prep for pilot course	CFNOS Instruction staff	NLT End Sep 08

Follow-up Action:

SER	ITEM	OPI	Estimated Due Date
1.	Draft Executive Summary of QSPWB	CPO2 Gracey	Mid Jul 08
2.	Staff ROP, Executive summary and QSP to DMTE IAW CFNOS SOs.	CPO2 Gracey	1 Aug 08
3.	Request MITE ID for QL4 OJPR (DMTE MASM) http://maritimeapp.mil.ca/dmte/mite/intro_e.asp?dmte=1	CPO2 Gracey	1 Aug 08
4.	Include the requirement for successful completion of the 1030 Math Exam in order to proceed to the QL 5A course. Needs to be included in the JBS/Career progression table.	COP2 Gracey	1 Sept 08
5.	Include the following changes into the QL3 QSPWB: • Modules 1-6 1030 Math Package • Critical EC for the 1030 Math Package Mod 1-6	CPO2 Gracey	Oct 08
6.	Schedule pilot Serials for QL4 and QL5A in MITE	CFNOS MITE	1 Sep 08
7.	Finalize QL4 QSP	Coordinator CFNOS standards and Instruction	or earlier End Dec 08
8.	Finalize QL5A QSP	cell CFNOS standards and Instruction	End Dec 08
9.	Equipment Familiarization Packages be put forth by the CCA for QL4 and QL5A	cell CFNOS Standards	End Dec 08
10.	Implement pilot course changes and feedback (QL4 and QL5A)	Cell CFNOS standards and Instruction	60 days post course
11.	QL3 QSP (Nov 08) – Implement recommendations from QL4 QSP Train/No train list	cell QL3 QSPWB/ DMTE 2	completion During respective Board

SER	ITEM	OPI	Estimated Due Date
12.	QL5B QSP (Jan 09) – Implement recommendations from QL5A QSP Train/No train list	QL5A QSPWB / DMTE 2	During respective Board

Other Considerations

73. During the last day, the board moved on to discuss equipment familiarization and Pre-instruction Package (EF) requirements for all training. The basic outline for the QL 4 and QL 5A EF packages looks as follows:

QL 4 EF/PIP

- Tour/Demo of Equipment, Spaces and Operating Software of:
- STIR, 57mm, ESSM, EO, IR. (To be completed onboard ship and/or NCOT); and
- SSD/CCS basic view of system and tour.
- Read applicable info in relation to applicable publications:
- NCPM 701 FC Drills and Procedures;
- CFCD 114H applicable pages; and
- CFCD 106 applicable pages.

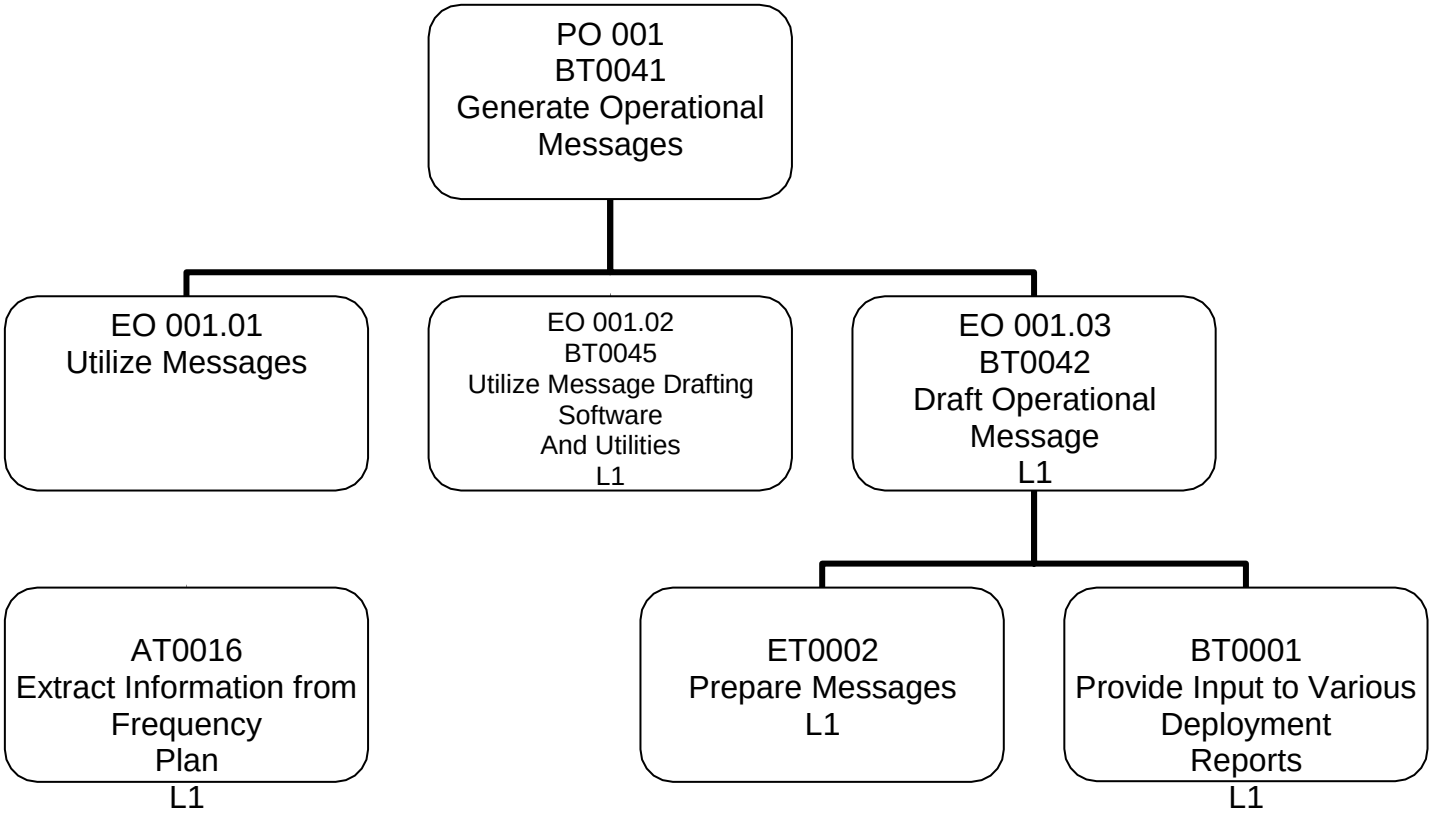
QL5 A EF/PIP

- Tour/Demo of Equipment, Spaces and Operating Software of:
- CIWS, RAMSES, SHIELD bridge module, CCS operation.
- Read applicable info in relation to applicable publications:
- NCPM 701 CIWS, SHIELD bridge module Drills and Procedures; and
- CFCD106/114H ref: EA,EP and EW/ELINT.

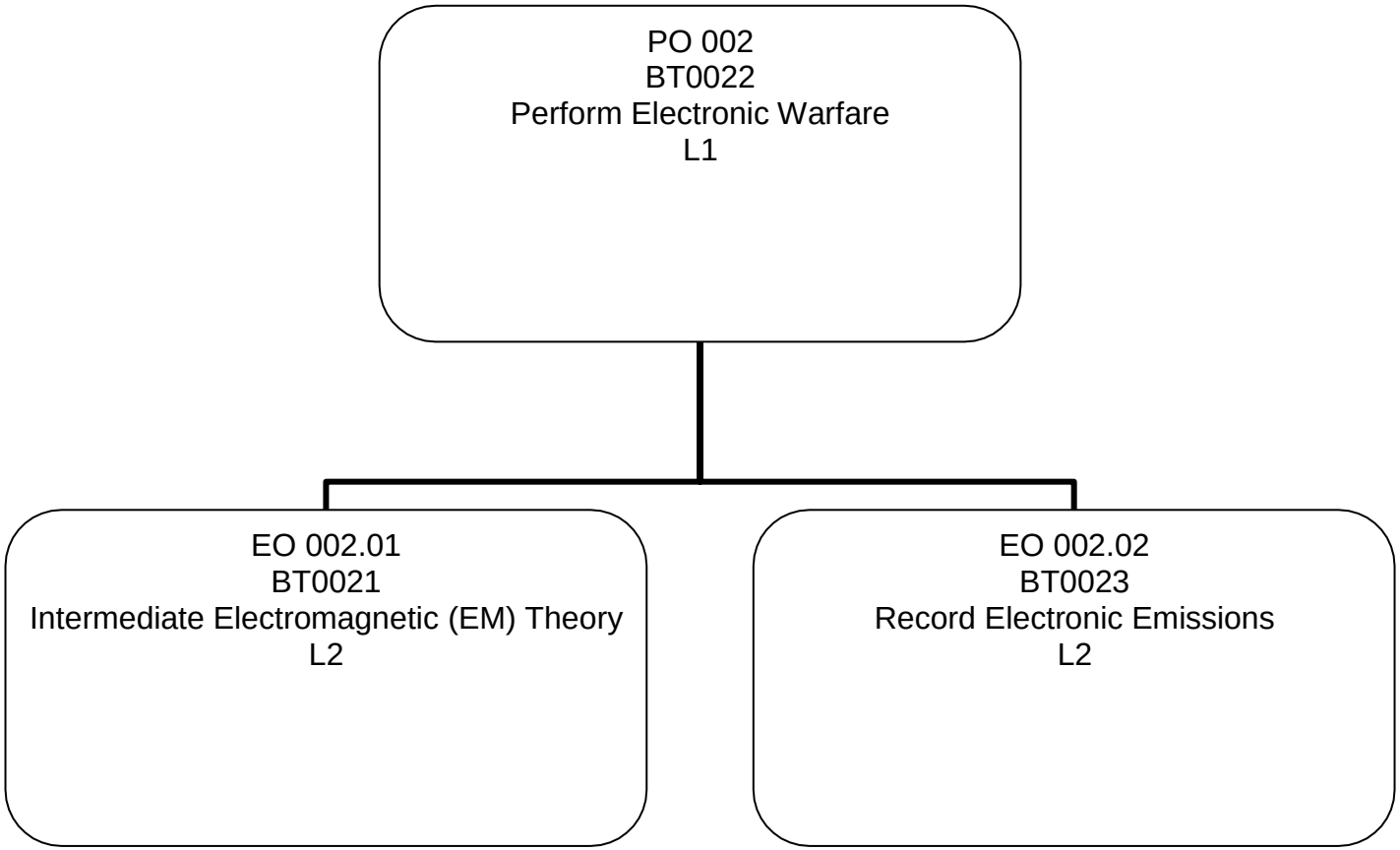
Conclusion

74. The Board Chair thanked members for their contributions during the QSPWB period and dissolved the board at 1200hrs 25 Jun 08.

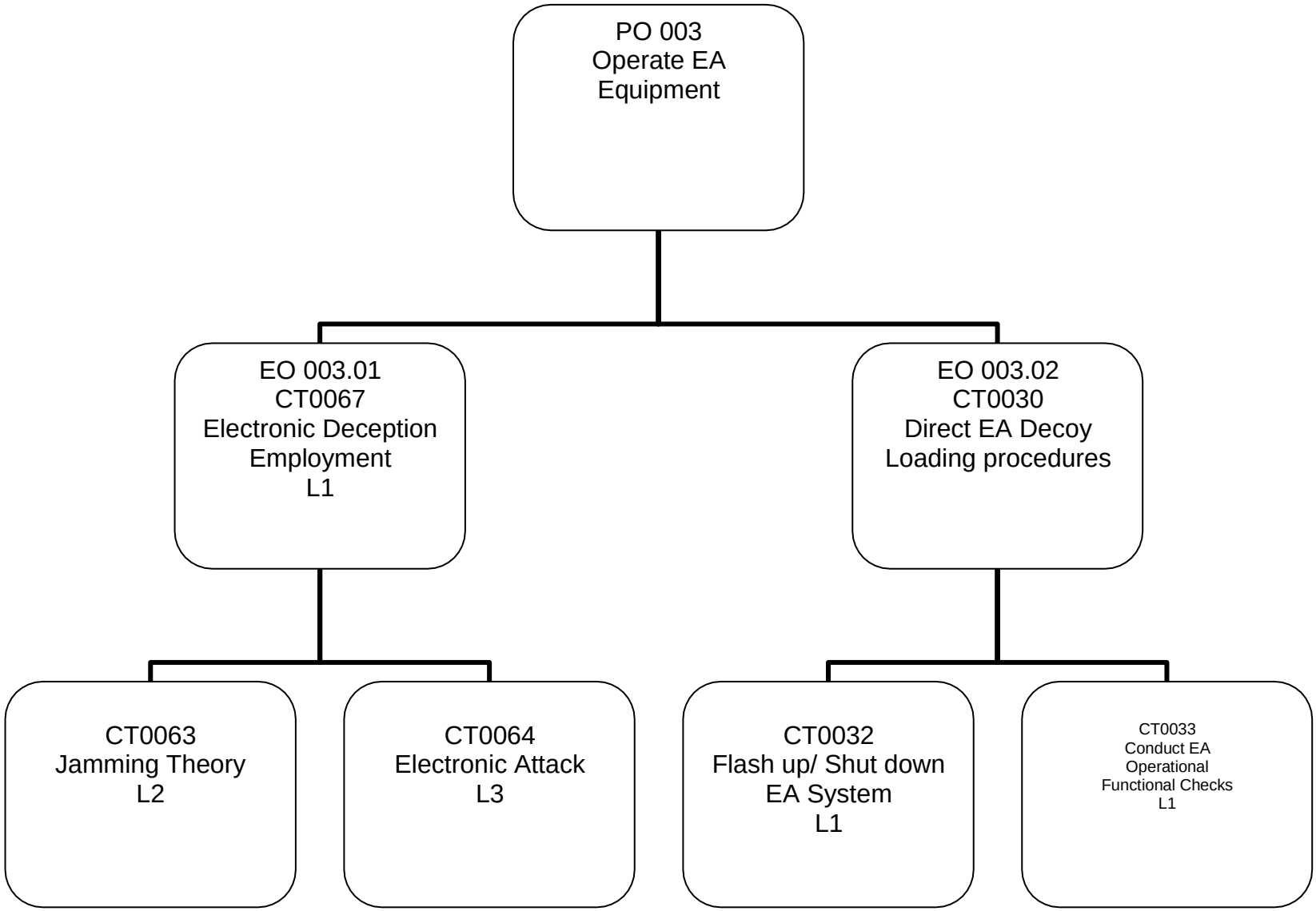
ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR



Approved: 20 Aug 2015
Version No:2.7
MITE Crse ID:120689



Approved: 20 Aug 2015
Version No:2.7
MITE Crse ID:120689



Approved: 20 Aug 2015
Version No:2.7
MITE Crse ID:120689

ANNEX E – ACRONYMS / ABBREVIATIONS

ABBREVIATION TERM/DEFINITION

AAW	Anti-Air Warfare
AOI	Area of Interest
AOP	Area of Probability
ASuW	Anti Surface Warfare
AWW	Above Water Warfare
CANEWS	Canadian Naval Electronic Warfare System
CBT	Computer Based Training
CCS	Command and Control System
CCOI	Critical Contact of Interest
CFMWC	Canadian Forces Maritime Warfare Centre
CFEWC	Canadian Forces Electronic Warfare Centre
CFNOS	Canadian Forces Naval Operations School
CIO	Communication Intercept Operator
CIS	Communication Intercept System
CIWS	Close In Weapon System
CMTU	Cartridge Magnetic Tape Unit
COI	Contact of Interest
COMM	Communication
COMSEC	Communication Security
CSE	Communication Security Establishment
CSTC	Combat Systems Training Centre
CW	Continuous Wave
DARO	Data Amplification Read-Out
DF	Direction Finding
DND	Department of National Defence
DTG	Date time group
EA	Electronic Attack
EC's	Enabling Check's
EF	Equipment Familiarization
ELINT	Electronic Intelligence
EMCON	Emission Control
EM	Electromagnetic
EO	Enabling Objective
EPM	Electronic Protective Measures
EP	Electronic Protection
ESM	Electronic Support Measures
ESR	Exercise Support Request
EW	Electronic Warfare
EWD	Electronic Warfare Director
FC	Fire Control
FCO(C)	Fire Control Operator CIWS
GHz	Giga Hertz
HFDF	High Frequency Direction Finding
HULTEC	Hull to Emitter Correlation
IAW	In Accordance With
ID Crit	Identification Criteria
IFF	Identification Friend or Foe
INTEL	Intelligence

IR	Infra-Red
IRCS	International Radio Call Sign
KHz	Kilo Hertz
LASER	Light Amplification by Simulated Emission of Radiation
LCP	Local Control Panel
LFMOP	Low Frequency Modulation on Pulse
LOB	Line of Bearing
LOF	Line of Fire
MARCORDS	Maritime Command Orders
MHz	Mega Hertz
MIPS	Maritime Intelligence Publications
MOP	Modulation on Pulse
NCOT	Naval Combat Operator Trainer
NEWC	Naval Electronic Warfare Centre
NEWT	NESOP Electronic Warfare Trainer
NSA	National Security Administration
NTDS	Naval Tactical Data Symbology
OCS	Officer Conducting Serial
OPGEN	Operation General
OPSTAT	Operational Status
OPTASK	Operational Tasking
ORTT	Operations Room Team Trainer
OTH	Over The Horizon
PC	Personal Computer
PC's	Performance Check's
PDW	Pulse Descriptor Word
PO	Performance Objective
PRB	Performance Review Board
PRF	Pulse Repetition Frequency
PRI	Pulse Repetition Interval
PSuM	Phalanx Surface Mode
PTA	Pulse Train Analysis
PTI	Phalanx Thermal Imager
QAB	Quick Action Button
QR	Quick Reaction
RADAR	Radio Detection and Ranging
RAMSES	Reprogrammable Advanced Multimode Shipborne ECM System
RCP	Remote Control Panel
RCS	Remote Control Station
RF	Radio Frequency
RFI	Request for Information
RFS	Request for Support
ROE	Rules of Engagement
RoW	Rest of World
SAC	Shipborne Air Controller
SAR	Synthetic Aperture Radar
SCONUM	Ship's Control Number
SDP	Shipboard Development Package
SDW	Signal Data Word
SHIELD	Shipboard Integrated Electrical Launcher Decoy system
SIF	Selective Identification Feature
SPT	Signal Processing Tool
SSD	Standard Shipboard Display
TACELINT	Tactical Electronic Intercept
TRB	Training Review Board
TRO	Track Read Out
XMT	Exempt

ANNEX F – SUGGESTED SEQUENCING AND TIMINGS

PO/EO	OBJECTIVE (EO / PO Title)	Classroom	Trainer	Demo	EC (Cr EC)	PC
EO 001.01	Utilize Messages	14		1		
EO 001.02	Utilize Message Drafting Software and Utilities	3		1		
EO 001.03	Draft operational Messages	22				
PC 001	Generate Operational Messages					**1**
EO 002.01	Intermediate Electromagnetic Theory	36			1	
EO 002.02	Record Electronic Emissions	3	13		8	
PC 002	Perform Electronic Warfare					**1**
EO 003.01	Optimize EW Sensor Employment	14			1	
EO 003.02	Direct EA Decoy Loading Procedures	13	8	2	8	
PC 003	Operate EA Equipment					**8**
		105	**21**	**4**	**18**	**10**

Total Number of Training Periods	158
Total Weeks of ATT	3.96

	Periods	Days
ATT Sub Total (Trainer, Classroom, Assessment etc)	158	19.75
TAT (Max 2 x 50 minute periods / week)	9	1.1
PT (3 x 50 minute periods / week)	11	1.4
BAT (Base Administration Time)	16	2
Course Duration Calculation	**194**	**24.25**

ANNEX G – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

Item	**Number of Days**	**QTY**
Trainer (NCOT, CSTC, Ship's Tours, GSF, Trinity)	3	
Classroom	17.25	
Trinity Tour		
PP&S		As required
PC Work Station		1 per student
Access to MCOIN Terminal		As required
References Classified/Unclassified		As required
Access to DWAN		1 per classroom
A/V Equipment (Projector)		1 per classroom
Instructor Work Station		1 per classroom
Calculator		1 per student
IT assets (Memory Storage Devices)		As required
Secure Lock-up		1 per classroom

ANNEX H – COSTING INFORMATION

Include information about cost in whichever format is applicable.
This annex is optional (**Not** to be deleted).

1. Estimates only, based on CF Cost Factors Manual FY 07/08
2. Student Salaries not included.

Course: NES Jnr Opr QL5		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Each year	Each succesive year	5 Year total
Airfare (Return)	6	$1,200.0		$7,200	$7,200	$36,000.00
R&Q	6	$35.0	69	$14,490	$14,490	$14,490.00
Incidentals	6	$17.5	69	$7,245	$7,245	$36,225.00
		subtotal		**$28,935**	**$28,935**	**$86,715.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S	12	$44.0	1	$528	$528	$2,640.00
Classsroom	1	$1,350.0	50	$67,500	$67,500	$337,500.00
Trainer	1	$1,350.0	31	$41,850	$41,850	$209,250.00
Computer requirements	12	$1,000.0	1	$12,000	$12,000	$60,000.00
DWAN connection	0	$0.0	1	$0	$0	$0.00
Instructional Materials	0	$0.0	1	$0	$0	$0.00
External Support	0	$0.0	1	$0	$0	$0.00
Other	0	$0.0	1	$0	$0	
		subtotal		**$121,878**	**$121,878**	**$609,390.00**
PYs	Quantity	Cost per day	# Days (2 Weeks Pre / post prep time)			
Instructor (P1)	1	$356.0	100	$35,600	$35,600	$178,000.00
Instructor (P2)	2	$329.0	100	$65,800	$65,800	$329,000.00
Intructor (MS)	0	$0.0	0	$0	$0	
Student	12	$184.0	94.9	$209,539	$209,539	$1,047,696.00
		subtotal		**$310,939**	**$310,939**	**$1,554,696.00**
	Total Estimated Costs			**$461,752**	**$461,752**	**$2,250,801.00**

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Version No:2.7
MITE Crse ID:120689

ANNEX I – WEEKLY STUDENT WRITE UP

Note: Insert additional annexes as required. Delete this annex if not required

Student Name	Date: (from/to)	Instructor Name:
Duties of Class Senior **-.50%** a) 2Nd failure to inform instructors of student/students; b) 2nd failure to ensure classroom is secure at end of day; and c) subsequent failures of duties listed above. **+.50%** a) Week without incident as course senior.		
Dress and deportment **-.25%** a) more than 3 improper reporting procedures to instructors in 1 week; b) failure to inform instructors of location of student/students; c) failure to ensure classroom is secured at end of day; and d) failure of following orders given by instructors. -.25% picked up during weekly morning inspection; -.50% picked up for 2nd infraction of same item that was previously noted and not satisfactory corrected; -1.0% picked up on CFNOS Divisions; +.50% students that for 3 consecutive weeks maintain the CFNOS requirements for standard of dress; and +1.0% student recognized by inspecting staff for having outstanding uniform ar CFNOS divisions.		

Instructor Signature:_____Date:_____

OPI Signature:_____ Date:_____

The OPI will put in archives on completion of course.

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Version No:2.7
MITE Crse ID:120689

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments are to be provided here.

16 Jun 2008 – CFFS (E). A QSPWB was held 26 May to 27 June 2008 (see ROP Annex C) at CFFSE (CFNOS is CCA, although the venue used was CFFSE). Provisionally approved performance Requirements from the Job-Based Specification were used in compiling the Train/No Train List. The focus of the board was primarily on training the operator to be sufficiently trained to effectively operate EW equipment, Weapon Systems (CIWS) and non-radar detection equipment. The Board decided that significant tasks and knowledge were better "Trained"/ developed using a shipboard development package (SDP) and that this should be designed based on the task list from Annex B through the occupation. See the ROP for more details.

4 July 2008;

The NES Op occupation and DMTE 2 decided in favour of a shipborne development package (SDP) over formal On-the-job training (OJPR), with the understanding the SDP could be tracked via a MITE Code (TBD) and ProgRep through the CMS/DMTE portal. The SDP will be administered, maintained and enforced by the NES Op occupation through the Training NCOs and Training Chiefs in each unit. It will be required by the occupation for core training but have no impact on timelines affecting promotion or course loading.
The impact of a SDP to develop expected higher proficiency levels that aren't covered in formal training under IT&E was realized and assessed as "Low Risk" before follow-on training.

24 June 2010:

The following amendments were authorized and implemented:
a) Delete Electro-Optic and Infra Red Detection System training at the QL-5A level and train at the QL-6A EWS level.
b) Delete CIWS training at the QL-5A level and train at the QL-4 level.

23 Jan 2012. The following amends were a result of a NESOP meeting in early 2011 and discussions on NESOP QL training with CFFS(E). These amend will reflect changes to teaching points to what is currently required in the fleet.
a. Delete EO002.03 Operate the SSD for EW/AWW Operations – taught at the QL6A level
b. Delete EO003.02 –Perform Pulse Train Analysis- School unable to do analysis never equip for only NRC OTTAWA all theory was moved to EO002.01
c. Delete EO003.03 – Record Electronic Emission – taught at the QL3 level
d. Delete EO003.04 – Perform Electronic deception -
e. Delete EO003.06 – Carry out EP Measures – not performed at this level
f. Delete EO003.07 – Execute Overt/Covert EW collection Operations

23 May 2012 – Administrative amendments to reflect the Command name change and the establishment of new lines of authority.

15 Jan 2013 – Changes were made to PC 001. The PC now requires student to write an EWX msg.
PC 002, was amended to say Written and time required was changed to 1 period.
Chapter 1 timing was changed as well as Annex F.

HCM Hybrid QSP Amendments:

These changes are designed to make this QSP HCM Hybrid compliant and ensure capabilities and limitations of new HCM driven equipment are covered.[The changes were fully discussed and drafted during the NESOP HCM QSP working group held 26 Apr -17 May 13. In attendance were;

PO1 M.P. Murray (CFNOS Standards)	PO2 J.E.J. Baker (CFNOS Insructor)
PO1 C.A. Macleod (Safe to Fire Cell)	PO2 R. Luce (CFNOS Instructor)
PO1 P.W.L. Tye (HMCS HALIFAX)	PO2 J.D. Dickie (HMCS FREDERICTON)

20 Aug 2015 – Changes made to Chapters 1 and 2 to reflect that the Authority is CNPTG. Secretarial changes made throughout to align with NPTORD. MITE ID and NQual Changed throughout QSP.

RQ-NCIOP-S1
Approval Sheet.pdf

30 Oct 2020 - Changes made to reflect RCN new Rank structure

ANNEX K – EVALUATTION MATRIX

EO/PO	EC/PC	PURPOSE	HOW ASSESSED	PASS/FAIL Criteria	TIME
EO 001.01	No Ec	To develop and test the students' ability to utilize Messages.	Formative: Developed through homework, end of lesson confirmation and tested in quizzes based on EO Teaching Points. Summative: Assessed as part of PC 001	N/A	N/A
EO 001.02	No EC - Practice	To develop the students knowledge of utilizing message drafting software	Formative: Developed through, end of lesson confirmation, practice from EO Teaching Points. Summative: Assessed as part of PC 001	N/A	N/A
EO 001.03	No EC	To develop the students ability to draft messages	Formative: Developed through, end of lesson confirmation and practice from EO Teaching Points. Summative: Assessed as part of PC 001	N/A	N/A
PO 001	**PC 001**	**To assess the students ability to Generate Operational Messages**	**PC 001 is a practical PC based on EO 001.01 and verbal confirmation questions taken from EO 001.02,and 001.03**	**Pass =80%**	**1 x 50 minute period**
EO 002.01	EC 002.01	To develop the students understanding of Electromagnetic Theory.	Formative: Developed through homework, end of lesson confirmation based on teaching points from EO 002.01. Summative: Assessed as part of PC 002	N/A	1 x 50 minute period
EO 002.02	EC 002.02	To develop the students ability to interpret SSD information	Formative: Developed through homework, end of lesson confirmation based on teaching points from EO 002.02. Summative: Assessed as part of PC 002	N/A	8 x 50 minute period
PO 002	**PC 002**	**To assess the students ability to Perform ELINT**	**PC 002 is a practical PC consisting of a checklist and verbal**	**Pass= 80%**	**12 x 50 minute periods**

		Collection SupervisionWW	**confirmation questions taken from all EOs in PO 002.**		
EO 003.01	EC 003.01	To develop the student the knowledge required to Utilize Electronic Deception	Formative: Developed through quizzes and tested in practical EC 03.01 based on scenario IAW Checklist from teaching points. Summative: Assessed as part of practical PC 003	Pass = 80%	1 x 50 minute period
EO 003.02	EC 003.02	To develop the students ability to direct EA Decoy Loading Procedures	Formative: Developed through practice and tested in Practical EC 03.02 based on Checklist and teaching points from EO 003.02 Summative: Assessed as part of PC 003	Pass = 80%	8 x 50 minute period
PO 003	**PC 003**	**To assess the students ability to Direct EA Decoy Loading Procedures**	**PC 003 is a practical PC based on the subject matter contained in all the EOs from PO 003 and will consist of the following:**	**Pass= 80%**	**8 x 50 minute periods (1 per student, includes debrief)**